

1 Introduction

Writing survey questions begins with defining a research objective; that is, what the researchers want to study and measure with the survey. For instance, the goal of the National Crime Victimization Survey (NCVS) is to measure the level of crime victimization in the United States. The next step is to identify constructs (i.e., elements of information) that are sought by the researcher (Groves et al., 2004). In the NCVS, one of the constructs to be measured is nonfatal personal and household property victimizations. Because constructs are theoretical in nature and tend to be broad and multidimensional, question writers need to identify dimensions of a construct (e.g., nonfatal personal crimes include rape or sexual assault, robbery, aggravated and simple assault, and personal larceny) and develop a taxonomy, reflecting the relationships among dimensions and subdimensions (e.g., larceny is a lesser theft than robbery). The goal of this process is to derive a list of single-dimensioned non-overlapping domains that can be operationalized with survey questions. Figure 1.1 presents a simplified example of how we move from a broad theoretical construct to specific indicators (without presenting the actual questions).

The next step is to identify what types of questions will be most appropriate, depending on the constructs or domains – opinions/knowledge about the types of crimes or experiences of crimes. Finally, we gather information on how respondents from our target population talk about these dimensions and subdimensions of the concepts of interest and using their vocabulary, translate the concepts into survey questions with respective response options. For example, focusing on the construct of identity theft, the NCVS asks questions for

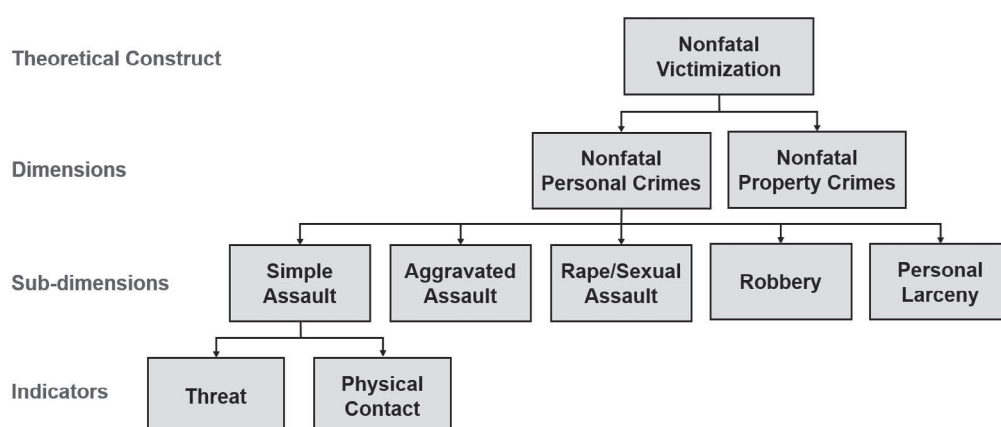


Figure 1.1 Taxonomy Example of Moving from Constructs to Specific Question Topics.

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each of the three dimensions defining identity theft – unauthorized use or attempted use of an existing account, unauthorized use or attempted use of personal information to open a new account, and misuse of personal information for a fraudulent purpose:

Since [DATE], have you or anyone in your household discovered that someone...

- a Used or attempted to use any existing credit cards or credit card numbers without permission to place charges on an account (Yes, No, Don't Know)
- b Used or attempted to use any existing accounts other than a credit card account – for example, a wireless telephone account, bank account or debit/check cards – without the account holder's permission to run up charges or to take money from accounts (Yes, No, Don't Know)
- c Used or attempted to use personal information without permission to obtain NEW credit cards or loans, run up debts, open other accounts, or otherwise commit theft, fraud or some other crime (Yes, No, Don't Know)

In the following sections, we introduce the typology of survey questions, briefly discuss the types of survey error, and focus on mechanisms that induce measurement error, given that this type of error is of most concern for questionnaire designers. The rest of this book discusses how to write various types of survey questions, focusing on design considerations motivated by the survey response model presented in [Chapter 2](#).

Types of Survey Questions

We distinguish among four types of survey questions that are discussed in detail in [Chapters 3–5](#):

- **Factual or behavioral questions** typically measure what happened, when it happened, and how often it happened. There is an objective truth associated with such questions even though it is rarely available to researchers. [Chapter 3](#) addresses how to write factual or behavioral questions.
- **Attitudinal questions** measure opinions and perceptions about an attitudinal object. There is no truth associated with such questions because they are subjective in nature. The lack of an objective standard makes it hard to detect measurement error in attitudinal questions. [Chapter 4](#) focuses on developing attitudinal questions.
- **Demographic questions** measure respondents' characteristics that may vary in terms of objective elements (e.g., age) or subjective elements (e.g., ethnicity and gender). [Chapter 5](#) includes guidelines and recommendations for writing demographic questions.
- **Knowledge questions** measure respondents' knowledge on a particular topic. There is an objective truth associated with these questions and it is known to researchers. Such questions are sometimes used to screen out respondents who do not have enough knowledge to answer follow-up attitudinal questions. Guidelines and recommendations on writing knowledge are discussed in [Chapter 5](#).

Another typology divides survey questions by question structure type:

- **Open-ended questions** do not present answer choices; respondents use their own words to provide a response.
- **Closed questions** provide respondents with a list of ordered or unordered answer choices. They can be multiple-choice questions (e.g., offering a frequency scale such

as *Always, Most of the Time, Half of the Time, Less than Half of the Time, Never*) or simple questions (e.g., offering Yes/No response options).

- **Partially closed questions** provide a list of ordered or unordered answer categories and also include an open-ended option at the end, for example, “*Other (Please Specify)*.”
- **Field-coded questions** are administered as open-ended, but interviewers code verbatim responses provided by respondents into one of a set of predefined categories.

Some questions are considered sensitive because they are potentially intrusive, are associated with perceived risk of disclosure to third parties, or invoke social desirability concerns. [Chapter 6](#) offers suggestions on how to write sensitive questions.

Survey Errors

Survey researchers are concerned with errors that can occur at each stage of the survey process. We define “error” as any deviation or departure from the desired outcome ([Groves et al., 2004](#)). [Groves et al. \(2004\)](#) present the survey lifecycle from a design perspective, demonstrating how a survey moves from the questionnaire design and data collection phases to survey statistics, adding a quality component that indicates where survey errors are likely to occur ([Figure 1.2](#)).

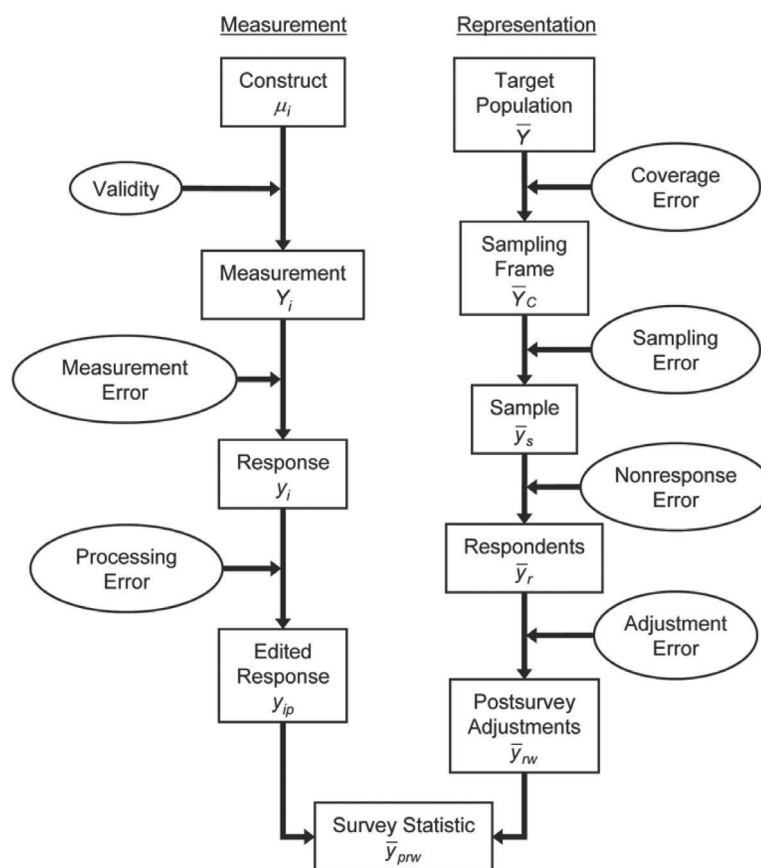


Figure 1.2 Survey Lifecycle from a Quality Perspective.

Groves et al. (2004) identify six main types of survey errors, classified as either errors of non-observation or errors of observation. The left-hand side of Figure 1.2 presents errors of observation (i.e., we have collected data from the sample member, but there are quality issues with the data collection process); the right-hand side presents errors of non-observation (we have failed to collect data from the sampled member, leading to quality issues). We briefly describe each of the errors below.

Errors of Non-observation

- Coverage error occurs because not everyone from the target population is represented on the sampling frame.¹ For instance, an address frame does not include homeless people, resulting in coverage error.
- Sampling error occurs because we are not conducting a census of the target population, but rather estimating a population parameter (such as mean or total) on a sample selected from the sampling frame. If we select a different sample, we would likely get a different estimate; thus, there is a sampling error associated with each survey statistic estimated from a sample.
- Nonresponse error is associated with failure to obtain responses from all eligible sampled units.
- Adjustment error occurs because of weighting or other adjustments to correct for sampling, nonresponse, or coverage errors.

Errors of Observation

- Validity reflects the gap between what a survey question intends to measure (i.e., measurement) and the theoretical construct it is supposed to measure.
- Measurement error occurs when answers provided by respondents to survey questions deviate from the truth. Going back to the example in Figure 1.1, we may discover that respondents intentionally misreport their true value on questions related to rape and sexual assault because the questions were too sensitive.
- Processing error is a result of any editing, coding, or imputation during data processing.

Each type of error has its own set of sources. This book is about writing survey questions; thus, our attention will focus on measurement error and the mechanisms that cause it.

Measurement Error

Various disciplines have slightly different perspectives on measurement error and potential fixes for it. Survey methodology combines the statistical view and the cognitive perspective. The statistical view on measurement error defines it as the discrepancy between the true and observed values:

$$y_{ij} = X_i + \epsilon_{ij}$$

where y_{ij} is the actual answer provided by respondent i to question j ;

X_i is the true value for respondent i ; and

ϵ_{ij} is the measurement error in the actual response provided by respondent i to question j .

The goal of statisticians is to measure ϵ_{ij} and adjust for it. They are interested in bias (the systematic effect of ϵ_{ij}) and variance (the variation of ϵ_{ij} across respondents) in a survey estimate. When we talk about bias, often we use terms such as *underreporting* when we underestimate the true value and *overreporting* when we overestimate the true value.

The cognitive view on measurement error is mostly concerned with the mechanisms that produce measurement error – for example, question misunderstanding, memory failure, and social desirability – and how to minimize their influences. We will adopt both views in this book, using them to lay the foundations for best practices and recommendations on writing survey questions with minimal measurement error.

Five main sources of measurement error are found in surveys: measurement error can occur because of the interviewer, the respondent, the mode of data collection, the survey questionnaire, or the language in which the questionnaire is administered. Measurement error because of the interviewer can be a result of observable (e.g., race and gender) or unobservable characteristics of interviewers (e.g., rapport with respondents) and interviewing techniques (e.g., pace of the interview). In fact, the survey literature is rich with examples of interviewer effects on survey estimates because of interviewer characteristics (e.g., Bailer et al., 1977; Fellegi, 1964; Rice, 1929; Schnell, 1997) or interviewing techniques (e.g., Hill & Hall, 1963; Mensch & Kandel, 1988; Olson & Peytchev, 2007; Weiss, 1968). For example, African American interviewers obtained lower levels of hostility and fear toward African Americans from white respondents than white interviewers (Schuman & Hatchett, 1976), and in a longitudinal study, Mensch and Kandel (1988) found lower reports of drug use when respondents were familiar with the interviewer. A large body of literature has demonstrated a significant component of variance because of interviewers for both sensitive and nonsensitive survey questions (e.g., Belak & Vehovar, 1995; Collins & Butcher, 1982; Davis & Scott, 1995; Fellegi, 1964; Gray, 1956; Groves, 1989; Groves & Magilavy, 1986; Hansen et al., 1960; Hanson & Marks, 1958; Kish, 1962; O’Muircheartaigh & Campanelli, 1998; O’Muircheartaigh & Wiggins, 1981; Schnell & Kreuter, 2005; Tucker, 1983), including self-administered items, when the interviewer is present in the room (e.g., Collins & Butcher, 1982; Groves & Magilavy, 1986; Mangione et al., 1992; O’Muircheartaigh & Campanelli, 1998).

Measurement error because of the respondent can be a result of respondent limitations or characteristics (e.g., cognitive ability, tendency to acquiesce, or provide socially desirable responses). Three chapters focus on questionnaire characteristics related to measurement error and ways to minimize the potential for it through deliberately manipulating the survey layout, controlling context effects, and minimizing question ambiguity and cognitive burden (Chapters 3–5). Chapter 6 discusses writing survey questions for two sensitive populations – older adults and children under the age of 18.

The mode of data collection has also been found to affect survey reporting, with self-administered modes yielding more accurate reports of sensitive behaviors than interviewer administration (e.g., Bradburn, 1983; Jones & Forrest, 1992; Miller et al., 1990; Schwarz et al., 1991a; Turner et al., 1992). However, there is puzzling evidence of interviewer effects in self-administered modes like audio computer-assisted self-interviewing when an interviewer is present (e.g., Hughes et al., 2002; O’Muircheartaigh & Campanelli, 1998; West & Peytcheva, 2014). We discuss questionnaire design for two self-administered modes (paper surveys in Chapter 7 and web and mobile web surveys in Chapter 8). Chapter 9 discusses designing questionnaires for studies mixing multiple modes of data collection.

Finally, the language in which a survey is administered impacts the survey response process (Peytcheva, 2020; Schwartz et al., 2010). To the extent to which language can cue socially desirable reporting through its link to cultural values and norms, it can induce measurement error. For example, one of the first studies that discovered the effect of language on survey responding was an experiment by Triandis et al. (1965), where Greek-English bilinguals were asked the same questions in both languages. The study found low correlation between responses provided by the same respondents when the questions differed in social desirability in the Greek and American cultures. Subsequent studies have demonstrated that language can be a powerful cue to the cultural frame that respondents adopt; they may provide completely different responses to the same questions, depending on the language used to ask those questions (e.g., Marian & Neisser, 2000; Marin et al., 1983; Peytcheva, 2019; Ross et al., 2002; Schrauf & Rubin, 2000; Trafimow et al., 1997). We discuss questionnaire design considerations for multilanguage surveys in Chapter 10.

Assessing Measurement Error

As with any type of survey error, we need to know the true value, in addition to the observed one, to be able to quantify the measurement error component. Administrative records, split ballot experiments, repeated measurements, and parallel or alternative measures (including physical measurements) have been used to estimate measurement bias. However, in most cases, estimation of bias is practically impossible. Thus, within one survey administration, we resort to proxy indicators for measurement error bias, such as item missingness, inconsistency of responses, context effects (including primacy and recency), mode differences, time on task, acquiescence,² and straightlining.³

Psychometricians use concepts such as validity and reliability to gauge the quality of their measures. In surveys, validity refers to the extent to which a survey question accurately measures the construct of interest, while reliability refers to variability of responses over repetitions. Chapter 13 describes how reliability and validity can be assessed through simple statistics (such as correlations and kappas) and complex modeling techniques (such as Multitrait-Multimethod Modeling and Item Response Theory).

Chapters 11 and 12 provide approaches commonly used by survey methodologists and question writers to evaluate and test survey questions. Chapter 14 discusses usability testing of survey instruments. These approaches are useful tools to reduce measurement error arising from questions and modes of data collection.

About This Book

This book is intended to be an introductory and practical textbook on questionnaire design. Our first objective is to present a broad overview of the existing literature, drawing on both theoretical and empirical research. Our second objective is to present practical principles, guidelines, and best practices for developing and testing survey questions.

This book consists of three themes. The first is the survey response process model (Schwartz et al., 1996; Tourangeau, 1984; Tourangeau et al., 2000), which will serve as the theoretical framework to establish the basic principles of writing different types of survey questions. These principles and best practices are supported by both keystone theoretical and experimental studies and the latest research. We present the cognitive processes related to answering survey questions (Chapter 2), followed by detailed discussion

on how to write different types of survey questions (Chapters 3-5), knowing the theoretical framework of the survey response model.

The second part of this book focuses on special topics such as asking sensitive questions (Chapter 6), developing questionnaires for older people and children (Chapter 6), designing paper surveys (Chapter 7), designing web surveys and optimizing the questionnaire for mobile devices (Chapter 8), developing questionnaires for a multimode survey (Chapter 9), and conducting surveys in multiple languages and cultures (Chapter 10).

The third theme of this book focuses on methods for testing and evaluating survey questions (Chapters 11-13). Topics include, but are not limited to, focus groups, cognitive interviewing, expert review, Questionnaire Appraisal System, behavior coding, respondent and interviewer debriefing, experimental pretest, and pilot studies. Given extensive web survey coverage, we also discuss usability testing for self-administered surveys (Chapter 14). The last chapter (Chapter 15) provides a brief overview of the use of artificial intelligence and machine learning for questionnaire development and evaluation.

Even though this book is about writing survey questions, invaluable sources are available that can be used to find questions that match one's research objective and can be used as is or adapted for one's own purposes. Below are some of the question banks that we recommend:

- <https://wwwn.cdc.gov/qbank/>
- <https://www.pewresearch.org/politics/question-search/>
- www.icpsr.umich.edu

Notes

- 1 A sampling frame is the listing of all units in the target population.
- 2 The tendency of respondents to agree with the question statement, regardless of how they actually feel.
- 3 The tendency of respondents to select the same response option for questions that use the same scale, especially when presented in a grid.

2 Survey Response Model

Introduction

According to the most influential conceptual framework of the survey response process (Sudman et al., 1996; Tourangeau, 1984; Tourangeau et al., 2000), respondents undergo four cognitive stages when answering a survey question (Figure 2.1). We will use a question from the Consumer Expenditure Surveys (CE) as an example to illustrate the four stages of the survey response process:

In the last 12 months, how much did (you/your household) spend on your last purchase of living room, family, or recreation room furniture?

The first stage is **comprehension**. Respondents must understand the meaning of the survey question as intended by the question writer to provide a meaningful answer. For instance, they have to figure out what counts as “living room, family, or recreation room furniture” and what the reference period (the last 12 months) covers. They also must understand the intent of the survey question. In this instance, the question writer wants to know the expenditure of the respondent’s *last* purchase of living room, family, or recreation room furniture in the last 12 months.

The second stage is **retrieval** of information. Respondents search for relevant information (say, all purchases of living room, family, or recreation room furniture that occurred in the last 12 months) in their long-term memory and move the retrieved information to their working memory. This stage follows the determination that the survey question asks about expenses of their very last purchase in the last 12 months.

The third stage is **judgment and estimation**. That is, respondents integrate the relevant retrieved information to form an attitude or provide an estimate related to a behavior when the retrieved information cannot be directly used as an answer. In our example, after respondents retrieve all episodes of living room, family, or recreation room furniture that occurred in the last 12 months, they will need to decide which episode represents the last purchase that occurred in the last 12 months and then retrieve the expenditure for that purchase. If the respondents happen to have a receipt of that last purchase and are motivated to do so, they could pull out the amount directly. If they do not have a receipt, they would need to come up with an estimate based on some strategy (e.g., the TV cannot cost more than \$500). As we can see from the amount of work required at the retrieval and judgment and estimation stages of the survey response model, this question incurs rather burdensome cognitive tasks that respondents have to engage in to provide an answer.

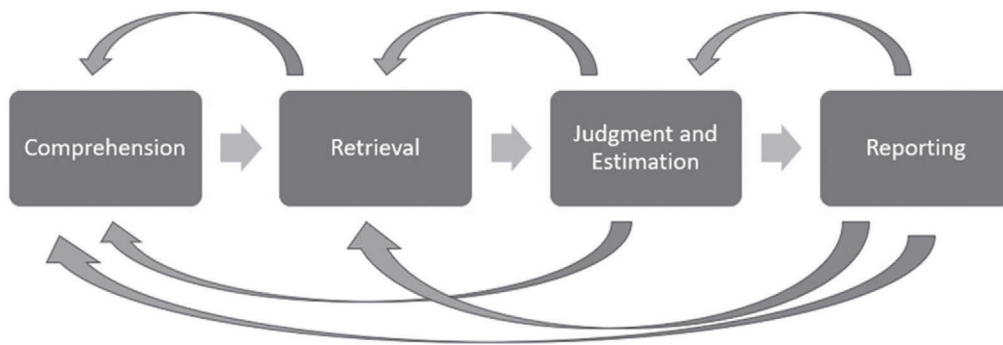


Figure 2.1 Survey Response Model.

At the fourth stage – **reporting** – respondents edit or map the resultant judgment or estimate to one of the provided response options. Our example question asks respondents to provide a dollar amount. The respondent could report the dollar amount retrieved from the receipt, their memory, their best guess, or an estimate based on some strategy they use. Sometimes, respondents are provided a list of response options and they have to map their estimate into one of the options. At this stage, survey respondents may choose to edit what they retrieved before reporting it, for different reasons. For instance, the respondent may not feel comfortable reporting that their TV costs more than \$1,000 and may decide to report that they spent \$500 on the TV instead.

Next, we discuss each stage in detail. It is important to note that survey respondents do not necessarily follow the four stages sequentially and conscientiously. It is common that respondents engage in satisficing behaviors by skipping a stage completely or taking a cognitive shortcut at a particular stage (e.g., [Krosnick, 1991, 1999](#)). Sometimes respondents may jump ahead (e.g., they start the retrieval stage before the interviewer finishes reading the question) or backtrack (e.g., as respondents are retrieving information, they may go back to the comprehension stage to adjust their understanding of the survey question).

Comprehension

Comprehension of a survey question occurs at three levels: syntactical, semantic, and pragmatic ([Tourangeau & Bradburn, 2010](#); [Tourangeau et al., 2000](#)). The **syntactical** process involves grammatical analysis of the question to identify its presupposition (What assumption does the question make?) and focus (What information is requested of respondents?). Survey questions with a complex syntactical structure are difficult for respondents to process and answer. Studies using eye-tracking and response times data have demonstrated that longer questions, questions with multiple clauses, negatively phrased questions, and syntactically ambiguous questions are more burdensome and take longer for respondents to understand and answer ([Just & Carpenter, 1993](#); [Kamoen et al., 2017](#); [Lenzer et al., 2011](#); [Yan, 2023](#); [Yan & Tourangeau, 2008](#)). As a result, question writers should avoid double-negative questions and double-barreled questions because of their overly complicated syntactical structure. Double-negative questions are

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questions with more than one negative word in the question stem and response options. An example double-negative question is shown below with the negative words bolded:

All in all, do you favor or **oppose** Congress passing a law **not** allowing any employer to force an employee to retire at any age? FAVOR, OPPOSE.

Double-negative questions also present an analytic problem when the negative end of a scale is endorsed because it is unclear what the respondent's position truly is. In our example, when a respondent selects "oppose," it is not clear whether opposing a law not allowing any employer to force an employee to retire is the same as favoring a law allowing employers to force employees to retire at any age.

Double-barreled questions are questions asking about more than one topic. An example double-barreled question is shown below with the double topics bolded. The analytical challenge with double-barreled questions is that, when a respondent answers "yes" to this question, it is not clear which of the two things (dealing effectively with AIDS or dealing with sexually transmitted disease) they believe in.

Do you believe that the U.S. health care system is dealing effectively **with AIDS and with sexually transmitted diseases**?

Another syntactical problem to be avoided is faulty or inappropriate presupposition carried by a survey item. For instance, a survey question asking about the number of hours respondents worked outside in their garden presupposes that respondents have a garden, which is unlikely to be correct for everyone.

Semantic comprehension of a survey question is understanding the meaning of each word in the question. There are several hurdles to semantic comprehension. First, technical terms (e.g., myocardial infarction) and idioms (e.g., to build a nest egg) are less likely to be understood uniformly by the general population and thus should be avoided. Second, some words have multiple meanings; for instance, the word "table" can refer to a coffee table or an Excel table. It can also be a verb as in "to table a conversation." Third, vague concepts (e.g., children, ill effects) and vague quantifiers (e.g., very often) mean different things to different people. For example, in a classic study, [Belson \(1981\)](#) showed that some of his respondents took the word "children" to mean young people in general regardless of relationship to themselves whereas others took the word to mean their offspring regardless of age. [Schaeffer \(1991\)](#) demonstrated that white respondents who reported experiencing an emotion "pretty often" actually experienced that emotion about the same number of times per month as black respondents who reported "very often." Fourth, even words commonly used in daily life

Suessbrick et al. (2000) asked respondents about their understanding of one question used in the Current Population Survey ("Have you smoked at least 100 cigarettes in your entire life?"). About half of their respondents (54%) understood the word "smoking" as including any puffs whether inhaled or not, which is the intended meaning of the question. However, the other half (46%) thought that smoking in this question referred only to puffs that one inhaled and provided an answer based on that misunderstanding.

could be understood in a different way from the one intended by the question writer (see text box example). Fifth, sometimes question concepts do not have a straightforward mapping to a respondent's life circumstances. For instance, the CE question about expenditure of the last purchase of living room, family, or recreation room furniture in the last 12 months applies to the purchase of an end table. However, it is not clear to people whether it also applies to the purchase of a floor lamp (Schober & Conrad, 1997; Schober et al., 2004).

Respondents also need to understand the intent of survey questions, which involves the determination of implicatures or the **pragmatic** meaning of the questions. To infer the pragmatic meaning, respondents resort to principles they engage in everyday conversations, treating a survey as conversational interaction (Conrad et al., 2014; Tourangeau et al., 2000). The philosopher H. Paul Grice introduced the Cooperative Principle as the basis of human interactions (Grice, 1975). The principle assumes that participants in a conversation cooperate with each other, providing information that is relevant (Maxim of Relation), truthful (Maxim of Quality), sufficient but not redundant (Maxim of Quantity), and clear (Maxim of Manner). We assume that survey respondents apply these maxims to make inferences about the intent behind survey questions, drawing on contextual cues and generating answers based on the inferred intent. For instance, respondents are found routinely using contextual cues (such as the survey name, sponsor, prior questions, or response options) to help make sense of survey questions because the context is relevant to the goal of question-answering based on the Maxim of Relation. For example, Galesic and Tourangeau (2007) demonstrated that respondents perceived the same behaviors as more likely to represent sexual harassment when the survey was presented as "Sexual Harassment Survey" than when the survey was called "Work Atmosphere Survey." Similarly, Strack et al. (1991) showed that students made sense of a fictitious law ("educational contribution") based on prior survey items. A random half of students were asked about US tuition whereas the other half were asked about Swedish financial support. Those who were first asked about US tuition inferred that the educational contribution law was about students paying money, whereas those who got the Swedish financial support question took the law to be about students receiving money. In this case, the interpretation is not intended by the question writer.

Retrieval

Retrieval involves searching for relevant information in long-term memory and moving the retrieved information to short-term or working memory. It is particularly relevant to answering behavioral questions, as shown in Chapter 3.

Characteristics of events affecting retrieval: time of occurrence, proximity to temporal boundaries, distinctiveness, and importance or emotional impact.

The quality and ease of the retrieval process are affected by characteristics of information being retrieved. First, the older the information, the harder it is to retrieve and the less detailed the retrieved information is (Jobe et al., 1993). Forgetting is a major cause of respondents' underreporting requested information. Second, mundane and frequent events are harder to recall than distinctive and salient events because of blending of memories and changes of memories over time. For instance, people

remember better and remember more details about the purchase of a house than a trip to a grocery store. Third, people tend to misdate when an event or behavior occurred, leading them to report an event or experience that occurred outside the reference period, a phenomenon known as telescoping (Neter & Waksberg, 1964). All these factors are relevant to setting the appropriate length of a reference period for questions asking about behaviors.

The example question from CE on expenditure of last purchase of living room, family, and recreation room furniture in the last 12 months is seemingly a cognitively difficult question because it asks people to recall a purchase that occurred during the last 12 months. However, furniture purchase is not a mundane event and is likely memorable. Still, respondents may misdate a purchase or retrieve expenditure for the wrong purchase.

Furthermore, people do not take in everything that happens in their lives. Some information is never noticed and never stored in long-term memory, and, consequently, cannot be recalled, regardless of how we ask about it. For example, Lee et al. (1999) showed empirically that parents did not encode the exact immunizations their children received and thus could not recall what immunizations their children received, regardless of whether they were asked immediately or 2 weeks after the visit and regardless of what recall cues were presented to them. Another encoding problem affecting retrieval is that information might be encoded into a different category or with a different label than what is asked in the survey question. For example, respondents may remember vacuuming three times a week, but they may not encode that piece of information as a “moderate-intensity leisure-time physical activity.” As a result, they would not necessarily recall that piece of information when answering the 2022 National Health Interview Survey that asks them how often they do “moderate-intensity leisure-time physical activities.” Question writers should be aware of both encoding issues when developing survey questions.

The context in which the survey questions are answered affects this stage. By context, we mean a variety of factors ranging from interview setting and interviewer characteristics to question order, visual layout of response scales, and so on. Chapters 3 and 4 have detailed discussions on the impact of context on survey answers, which are also called context effects. Chapters 7 and 8 also talk about how to guard against context effects in paper and web surveys. At this stage, context affects retrieval by serving as a retrieval cue or by priming certain information for more accessible retrieval. For instance, researchers have found that the correlation of self-reported measures of a specific domain (say, marriage satisfaction) and a general domain (life satisfaction) is higher when the specific question is asked before the general question than when the general question precedes the specific question (Schwarz et al., 1991b; Strack et al., 1988; Tourangeau et al., 1991a, 1991b; also see Mason et al., 1994). This is because answering the specific question on marriage satisfaction has made information about marriage highly accessible in working memory. Because marriage is an important part of life, highly accessible information about marriage is also used to form a judgment about life satisfaction, yielding a high correlation between the two self-reported measures.

In general, providing respondents retrieval cues has been found to improve recall. A retrieval cue is a prompt to memory and provides clues that help with search of long-term memory. Chapter 3 discusses strategies to improve recall, including providing retrieval cues and using landmark events to set a reference period.

Judgment and Estimation

Sometimes the retrieved information can be directly used as an answer to a survey question. This happens when respondents have an existing attitude or a running tally of events. For instance, people

Strategies survey respondents adopt to derive an estimate or a judgment: bottom-up, top-down, and impression-based.

typically can retrieve their answers directly from their memory when asked how many houses they have purchased or how they feel about broccoli. However, in the absence of an existing attitude or an estimate, respondents will have to form a judgment or derive an estimate on the spot by applying one of the following estimation strategies to integrate the retrieved information: bottom-up, top-down, or impression-based (Tourangeau, 2018; Tourangeau & Bradburn, 2010; Tourangeau et al., 2000).

With the bottom-up strategy, respondents use specific retrieved information to add up to an estimate or to derive a judgment. To answer a survey question on the number of dental visits in the last 12 months, for instance, respondents add up specific episodes of dental visits recalled (e.g., “I had dental cleaning in February, and I also had two cavities filled in August.”) to yield an estimate (“2”). Similarly, when using the bottom-up strategy to generate answers to attitudinal questions, respondents combine specific beliefs or feelings retrieved about a political candidate (e.g., “This candidate supported abortion.”) to derive an overall attitude about the candidate (“This candidate must be very liberal”). Also see Tourangeau (2021) for a discussion of the belief-sampling model of forming an attitude on the fly and Chapter 4 on attitudinal questions.

By contrast, with the top-down strategy, respondents project general information (e.g., a rate or a general value) to the specific situation requested by the survey question. For instance, when answering the same question on the number of dental visits in the last 12 months, respondents may apply a retrieved rate (e.g., “twice a year”) to the last 12 months to derive an answer (“2”) regardless of what actually happened during that period of time. In the same way, respondents may use a retrieved general value (e.g., “Democrat politicians are liberal”) to derive their answer to the question about a particular Democrat candidate.

The top-down strategy could be used as an impression-based method, which is often used when all respondents could retrieve is some sort of impression or stereotype. For instance, respondents apply an impression (“I never have dental problems”) to yield an answer of zero to the question about dental visits in the last 12 months, and a stereotype (“All politicians are corrupt”) to give a negative rating to a particular political candidate.

For questions about behavioral frequency, the survey literature shows that the accuracy of self-reported behavioral frequencies is best when respondents can directly retrieve an answer from their long-term memory and worst when respondents adopt an impression-based strategy (Burton & Blair, 1991). Characteristics of the behavior, such as saliency or importance, frequency, and regularity of the behavior, play a role in respondents’ choice of the bottom-up or top-down strategy (Brown, 2002; Sudman et al., 1996). The bottom-up strategy tends to underestimate self-reported behavioral frequency measures because retrieval of specific episodes is likely to be incomplete (Burton & Blair, 1991). The top-up strategy is more likely to overestimate self-reported behaviors because of exceptions to the rate and rate misestimation (Burton & Blair, 1991).

Furthermore, context affects judgment and estimation by suggesting a norm or a standard of comparison to be used as the basis for forming a judgment. For instance, Couper and colleagues (2007) asked web respondents to rate their health. Some of respondents were shown a picture of a healthy woman jogging, while others were presented with an image of a sick woman lying in a hospital bed. Respondents were found to rate themselves healthier when presented with the picture of the sick woman than the healthy woman. This is because respondents used the picture as a basis for comparison, leading to a contrast effect.

Reporting

The fourth step is reporting the judgment or estimate derived at the end of the judgment and estimation stage. Closed questions offer respondents a list of response categories. Ideally, respondents can map the recalled or derived judgment or estimate to one of the response options provided to them. Exact match of this type is desired to ensure validity and reliability of self-report measures. However, mapping is not always an easy or straightforward process.

First, respondents sometimes deliberately do *not* choose the response option that best matches their answer. This happens with two types of survey items. When asked a sensitive question invoking social desirability concerns, respondents edit their answer before reporting it to avoid embarrassment or disclosure of behaviors or attitudes that do not conform to societal norms (Tourangeau & Yan, 2007; Yan, 2021; Yan & Cantor, 2019). Chapter 6 discusses strategies for asking sensitive questions.

Sometimes surveys use gate questions to determine whether follow-up questions are applicable to respondents. Respondents are found to deliberately choose a response option that will disqualify them or their household from being eligible and, thus, avoid additional follow-up questions, regardless of whether that selected response option reflects their true opinion or situation. This is called motivated misreporting. To reduce motivated misreporting to gate items, question writers should employ a grouped question format (that is, asking all gate questions together first before asking follow-up questions) instead of the interleaved format (asking the first gate question, followed by follow-up questions, and then the subsequent gate questions, followed by follow-up questions) (e.g., Daikeler et al., 2022; Eckman et al., 2014).

Second, formal features of response options affect respondents' choice of a response. We discuss five formal features below as an example and caution that question writers should pay extreme attention to such features when designing a response scale. Response order effects refer to the effect of the order in which response options are presented on resultant answers. When a list of unordered options is presented in a self-administered survey, options displayed at the beginning of the list are more likely to be selected, resulting in primacy effects (Galesic et al., 2008; Krosnick & Alwin, 1987). However, when the same list is read to respondents in interviewer-administered surveys, options toward the end of the list are more likely to be selected because they are heard last, leading to recency effects (Holbrook et al., 2007). For response scales comprising a list of ordered options (e.g., a Likert scale running from Strongly Agree to Strongly Disagree), scale points closer to the start of a

Formal features of response options affecting answers include order of response options, range of response options, numeric labels, colors and shade, and visual layout.

scale are more likely to be selected, regardless of the mode of data collection, producing scale direction effects (Keusch & Yan, 2018, 2019; Yan & Keusch, 2015), especially for longer scales and complex survey questions (Yan et al., 2018).

Another formal feature of response options – the range of response options – is sometimes used as a frame of reference by respondents to estimate their own or other people's behavioral frequencies. Respondents assume that the middle category of a frequency scale reflects “typical” or “average” behavior frequencies, whereas the ends of the scale reflect the extremes of the distribution. Respondents then use this information to estimate how much they themselves engage in a behavior in comparison to the “average” frequency. For example, Tourangeau and Smith (1996) showed that respondents reported having more sexual partners when they were given a high-frequency scale than when they were given a low-frequency scale. See Chapter 3 for more discussion.

In addition, the numeric labels attached to scale points are found to affect respondents' interpretation of the scale, and subsequently, the selection of a scale point. For instance, for a scale with the same number of points and the same verbal labels, respondents are less likely to choose scale points when a negative number is attached to them than when a positive number is used (Schwarz et al., 1991a; Yan, 2006; more discussion in Chapter 4). In a similar fashion, the color and shade of response options on a screen (Tourangeau et al., 2007) and the graphical layout of the response options (Lenzner & Höhne, 2021; Schwarz et al., 1998; Smith, 1993) also affect respondents' understanding and use of the scale. Respondents tend to see response options with similar color or shade as conceptually closer than when the same response options have different colors or shades (Tourangeau et al., 2007). Furthermore, respondents take the pyramid shape of a response scale to suggest that more people should be represented at the bottom of the scale than at the top of the scale (Lenzner & Höhne, 2021; Schwarz et al., 1998; Smith, 1993). Subsequent chapters on designing paper surveys (Chapter 7) and web surveys (Chapter 8) discuss the importance of visual characteristics of survey questions.

A third problem that could potentially occur at the reporting stage is a tendency to select an answer in ways independent of question content and the judgment or estimate derived at the end of the judgment and estimation step. In practice, this means that no matter what the question is asking, respondents will always select a particular answer category of a scale. This is known as response style (Van Vaerenbergh & Thomas, 2013). Well-known examples of response styles include straightlining (the tendency to select the same response option to a battery of items using the same set of response categories), acquiescence (the tendency to agree regardless of what the question is about), extreme response style (the tendency to select the end points of a response scale), and midpoint response style (the tendency to select the middle option of a response scale).

Summary

The survey literature has offered many general guidelines for writing good questions. We conclude this chapter with advice offered by Krosnick and Presser (2010). Similar conventional wisdom can also be found in Converse and Presser (1986), Schaeffer and Presser (2003), and Bradburn et al. (2004).

Advice on writing good questions:

- 1 Use simple, familiar words (avoid technical terms, jargon, and slang)
- 2 Use simple syntax

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- 3 Avoid words with ambiguous meanings (i.e., aim for wording that all respondents will interpret in the same way)
- 4 Strive for wording that is specific and concrete (as opposed to general and abstract)
- 5 Make response options exhaustive and mutually exclusive
- 6 Avoid leading or loaded questions that push respondents toward an answer
- 7 Ask about one thing at a time (avoid double-barreled questions)
- 8 Avoid questions with single or double negations

Advice on grouping questions into a questionnaire:

- 1 Early questions should be easy and pleasant to answer and should build rapport between the respondent and the interviewer, if administered in a telephone or face-to-face mode.
- 2 Questions at the beginning of a questionnaire should be engaging and explicitly address the topic of the survey, as it was described to the respondent prior to the interview.
- 3 Questions on the same topic should be grouped together.
- 4 Questions on the same topic should proceed from general to specific.
- 5 Questions on sensitive topics that might make respondents uncomfortable should be placed toward the end of the questionnaire.
- 6 Filter questions should be included to avoid asking respondents questions that do not apply to them.

Exercises

- 1 Describe the survey response model.
- 2 For each stage of the survey response model, describe at least two problems respondents may have. Also, which problems have you run into before?
- 3 This chapter includes various best practices and guidelines for writing good questions. Which guidelines/best practices you have heard before and which ones are new to you?

3 Writing Behavioral Questions

Introduction

Behavioral questions collect information on experiences and events that occurred in respondents' life. Such questions typically ask about what happened, when, where, how often, how long, how many times, or how much. In contrast to attitudinal questions (to be discussed in [Chapter 4](#)), behavioral questions are linked to an objective truth, at least in theory. However, the objective truth is often not known or may not be accessible to researchers. For instance, number of doctor visits can be verified through respondents' medical records; however, the records may have their own errors or not be available to researchers. Our discussion of developing behavioral questions will focus on the retrieval stage of the survey response model (see [Chapter 2](#) for a description of the survey response model). Because retrieval is the recall of information from long-term memory, we begin this chapter with a brief discussion on human memory and organization of autobiographical events (events that happened in a respondent's life) followed by a discussion of factors that might affect recall and practical advice on how to improve it.

Autobiographical Events and Human Memory

Knowledge of events may exist at different levels – it can be rather abstract for basic-level, daily events, such as eating breakfast ([Rosch, 1978](#)); generic, for repeated events that follow a particular script (e.g., doctor visits generally include talking to a receptionist, seeing a nurse, and then seeing a doctor); or very specific, for personal events that are of particular importance to respondents (e.g., high school graduation party). Thus, autobiographical memory consists of events of different levels of specificity that are likely to be recalled with different levels of accuracy. Several theoretical models describe the structure and links across autobiographical events (e.g., [Conway, 1996](#); [Kolodner, 1985](#); [Tulving, 1983](#)). What they all have in common is that they place autobiographical events in episodic memory, which along with semantic memory (knowledge about general events) and procedural memory (knowledge about how to do things) are systems in humans' long-term memory. The models differ in how they envision the retrieval process itself – from general to more specific memories (top down); sequential, allowing for chronological sequencing within a theme; parallel, based on relationships among events or specific episodes across themes, or a combination of all.

How autobiographical events are organized in human memory is relevant to how we structure survey questions and how we present retrieval cues to respondents. For example, experimental research has shown that for specific events, questions that ask about

what happened yield more accurate reports than questions about where the event took place or date of the event (Brewer, 1988; Wagenaar, 1986). We do expect that the ability to recall details and the accuracy of recall will vary by event type, as will other factors likely to affect recall and strategies to improve it.

Event Characteristics Affecting Recall and Strategies for Improving Recall

Event Characteristics Affecting Recall

As mentioned in [Chapter 2](#), multiple factors affect a respondent's ability to retrieve events directly and influence the extent to which inferences are reported rather than the events themselves. Events that are important to respondents, particularly unique events or those that are emotional at the time of occurrence, are remembered more accurately than mundane events, regardless of the elapsed time (e.g., [Linton, 1982](#); [Mathiowetz & Duncan, 1988](#)). In a self-study of memory, [Wagenaar \(1986\)](#) reported that the pleasantness of an event (as recorded in a diary over a 6-year period) was linked to the ability to recall the event a few years later. Similarly, [Thompson et al. \(1996\)](#) reported that events associated with strong emotions (both pleasant and unpleasant) were remembered better than neutral events.

In contrast, similar events require the spread of memory activation across multiple pathways (e.g., from general themes to more specific subthemes and isolated episodes); thus, little activation reaches each specific experience, and multiple events of the same type are easy to recall relative to unique events. Indeed, several early studies examining survey recall report that important and memorable events are reported better in surveys – for example, major purchases are reported more accurately than smaller ones ([Sudman & Bradburn, 1973](#)), and longer hospital stays are reported more accurately than shorter ones ([Cannell & Fowler, 1965](#); [Cannell et al., 1981](#)).

Strategies for Improving Recall

Landmark Events and Other Retrieval Cues.

Studies have demonstrated that even though respondents may not know the dates of events they are asked to report, they can report accurately dates for important events (landmark events, which are usually time-tagged) and can use them to put dates around less important events. Landmark events can be personal (such as college graduation, giving birth), or public (e.g., Thanksgiving, Christmas). Studies have demonstrated that highly salient landmark events yield better estimates of an event's date than other recall strategies (e.g., [Burt, 1992](#); [Thompson et al., 1996](#)).

The Neighborhood History Calendar, a method designed to collect event histories of community changes over time, contained memorable public events at different levels to help respondents with their recall task. Landmark events included national (e.g., the deposition of the king), regional (e.g., natural disaster), and local events (e.g., household electrification) ([Axinn et al. 1997](#)).

Decomposition. Asking multiple questions and breaking the recall task into smaller subtasks improves the probability of more accurate recall ([Means et al., 1989](#)). Question decomposition helps disentangle generic memories into specific events by asking

41a. SQATTACKHOW

(Other than any incidents already mentioned,) has anyone attacked or threatened you in any of these ways -

(Exclude telephone threats) -

Read each category.

(a) With any weapon, for instance, a gun or knife -

(b) With anything like a baseball bat, frying pan, scissors, or stick -

(c) By something thrown, such as a rock or bottle -

(d) Include any grabbing, punching, or choking,

(e) Any rape, attempted rape or other type of sexual attack -

(f) Any face to face threats -

OR

(g) Any attack or threat or use of force by anyone at all? Please mention it even if you are not certain it was a crime.

Ask only if necessary

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1 ☐ Yes - ASK 41b

2 ☐ No - SKIP to 42a

Did any incidents of this type happen to you?

Figure 3.1 Decomposition Example from the 2023 National Crime Victimization Survey Screener Questionnaire.

multiple questions rather than a single one. Figure 3.1 provides an example from the National Crime Victimization Survey (NCVS); instead of asking about assaults that respondents might have experienced in the past 6 months, the survey asks if someone has attacked or threatened the respondent by grabbing, punching, or choking them; using any weapon; or sexually attacking them. Such a strategy is used to avoid premature termination of memory search when respondents may feel they have nothing to report.

Asking multiple questions related to the event of interest also improves the probability an event will be recalled and will stimulate associations across questions. For example, instead of just asking about seatbelt wearing during the respondent's most recent car ride, multiple questions related to when was the last time the respondent drove or rode in a motor vehicle, where the trip began and ended, how long was the trip, whether the respondent was the driver or a rider and if so, where they sat in the vehicle can be asked to improve recall to the seatbelt question (Dillman, 2000).

Event History Calendars. Event history calendars (EHCs) have been used to improve recall in surveys for events that occurred in previous years or respondent's lifetime (e.g., Belli et al., 2001; Freedman et al., 1988). EHCs aid respondent recall through processes that capitalize on the sequential and hierarchical storage of memories, and their link to personal landmark events serves as an inherent cue. Historically, EHCs have been used in interviewer-administered surveys, but with the widespread use of web surveys, they are becoming integrated into self-administered surveys as well (e.g., National Survey of Family Growth; NSFG). Figure 3.2 presents an example from the paper EHC for NSFG 2017–2019. To facilitate monthly recall of sexual activity and the use of contraceptive methods

National Survey of Family Growth: Life History Calendar																																																	
	2014												2015												2016												2017												
Before 2014	Ja	Fe	Ma	Ap	My	Jn	Jl	Au	Sp	Oc	Nv	Dc	Ja	Fe	Ma	Ap	My	Jn	Jl	Au	Sp	Oc	Nv	Dc	Ja	Fe	Ma	Ap	My	Jn	Jl	Au	Sp	Oc	Nv	Dc	Ja	Fe	Ma	Ap	My	Jn	Jl	Au	Sp	Oc	Nv	Dc	
Your Age																																																	
Education																																																	
Births & Other Pregnancies																																																	
Marriages, Cohabes, Partners																																																	
Intercourse																																																	
Birth Control Methods																																																	

Your Date of Birth: _____ Date of First Intercourse: _____

Birth or Pregnancy Ending Dates:

1st: _____ 2nd: _____ 3rd: _____ 4th: _____ 5th: _____

Figure 3.2 National Survey of Family Growth Paper Life History Calendar, NSFG 2017–2019.

over the past 4 years, respondents are asked to record events of personal importance (e.g., marriages, pregnancies, and births) that would facilitate recall of those monthly events and guide the judgment of whether they occurred within the time frame of interest.

Use of Diaries. Asking respondents to keep a diary for a short period of time and record specific events is a well-known practice for frequent, nonsalient events. This approach minimizes the reliance on recall because respondents are asked to record the event of interest immediately after its occurrence (e.g., daily food purchases). Despite their promise, diaries do not always provide a good solution – one of the main concerns is that keeping a diary may unexpectedly change the behavior under study; other concerns are related to timely completion and keeping respondents motivated to complete the diary over time.

Impact of Time Since Event on Recall and Strategies for Improving Recall

Impact of Time Since Event on Recall

Respondents are often asked to report on behaviors or events that were experienced in the past rather than at the time of the interview and typically, a time frame (reference period) is provided within the question:

*Did you see a pediatrician **within the last 6 months**, that is since January 25, 2024?*

The accuracy of such responses often depends on the time lapse since the event – the longer the time lapse between the event of interest and the survey interview time, the less likely respondents are to remember it accurately, if at all (see discussion in [Chapter 2](#)). Some of the theories on forgetting suggest that this is the result of respondents experiencing similar events in between and those additional experiences interfering with the memory

for the original event (Gillund & Shiffrin, 1984; Johnson, 1983). Indeed, validation studies demonstrate decrease in recall accuracy with increase of time since the event (e.g., Cannell et al., 1981; Loftus et al., 1992; Smith & Jobe, 1994).

Proximity to the reference period boundaries is another factor that interacts with length of the reference period and can contribute to a respondent's uncertainty about whether an event took place within those boundaries. Respondents can mistakenly report events that happened before the reference period or omit events as if they happened outside of the reference period. The first error is called forward telescoping (events are wrongfully telescoped within the reference period), while the second is called backward telescoping (events from inside the reference period are mistakenly taken out and not reported). Backward telescoping is less of an issue because respondents generally have less uncertainty about more recent events. Telescoping errors were first reported by Neter and Waksberg (1964) in a longitudinal study of expenditure, home repairs, and alterations. The study noted forward temporal displacement of both expenditure amounts and number of jobs and that was particularly true for larger expenditures and jobs (e.g., buying a sofa vs. paint for lawn furniture).

Strategies for Improving Recall

Short Reference Period. In addition to the use of landmark events and EHCs to help facilitate recall and reduce telescoping, shortening the reference period as much as possible given the behavioral frequency of an event might be a simple solution. For example, one of NCVS's objective is to produce annual estimates of crime victimizations. However, respondents are asked to recall and report crimes they might have experienced in the past 6 months instead of in the past 12 months. NCVS then uses the survey response data to extrapolate and produce annual estimates. Many studies manipulating the length of the reference period report a decrease in the number of events per unit time with length increase (e.g., meta-analysis by Sudman & Bradburn, 1973). This is likely the result not only of forgetting but also the increase in the total number of events that need to be recalled (Neter & Waksberg, 1964).

Bounded Recall. Bounded interviewing (providing a list of previously reported behaviors at the next wave of data collection) is a strategy commonly used in longitudinal surveys to reduce telescoping and improve recall. Neter and Waksberg (1964) compared the bounded interviewing approach to unbounded (simply asking respondents to report on jobs and expenditures within the reference period) and found that bounded recall significantly reduced telescoping by discouraging respondents to consider previously reported events. Cross-sectional studies have also demonstrated the utility of bounded recall in reduction of telescoping. Sudman et al. (1984) first asked respondents about events in the preceding month, then about events in the current month (which was of interest) and found that fewer events were reported in the current month when the interview was bounded; Loftus and Marburger (1983) used a landmark event (the eruption of Mt. St. Helens for respondents in Washington state) as a boundary of the reference frame to reduce telescoping.

Time on Task. Early investigations on strategies to improve recall have demonstrated that providing respondents more time to recall events improves response accuracy (e.g., Wagenaar, 1986; Williams & Hollan, 1981). Reiser et al. (1985) showed that it took several seconds to recall a specific event in response to a request, and the more difficult the retrieval task the longer it took. Allowing respondents more time to answer survey questions increases the accuracy of responses and at the question level, this strategy translates into asking longer questions. For example, asking questions with redundant wording,

thus providing respondents with more time to recall (Cannell et al., 1977; Cannell et al., 1981), or giving respondents longer deadlines to report back (Burton & Blair, 1991) have both yielded more accurate responses. Burton and Blair (1991) manipulated response time by providing respondents with the following instructions:

The next question is very important. After I finish reading the question, I would like you to spend at least fifteen seconds thinking about it. I will let you know when the fifteen seconds are up. If you wish to take more time, just let me know. Okay?

Overall Survey Length. Several studies have demonstrated that as surveys become longer, respondents' motivation deteriorates and willingness to work on retrieval decreases over time (e.g., Galesic & Bosnjak, 2009; Peytchev, 2007). Such findings suggest that the most important behavioral questions should be placed earlier in the questionnaire, and likely that behavioral questions should precede attitudinal ones when logically possible.

Estimation Strategies for Answering Behavioral Frequency Questions

Besides recall, the use of an estimation strategy to answer questions about frequency of a certain behavior affects the reported answers. Chapter 2 describes three estimation strategies respondents employ to come up with the number of times events happened in the past. With the bottom-up strategy, respondent enumerate every single instance of the event of interest and add them up to yield an estimate. The top-down strategy employs a rate-based estimation, when respondents apply a rule to come up with a response that is not an exact count of events, typically used when information about a "usual" or very frequent behavior is asked or when the reference period is so large that it does not allow for individual events to be recalled. An extreme form of the rate-based estimation is the impression-based method, where respondents retrieve an impression (e.g., I always wear my seat belt) and use it to form an answer (number of times the seat belt is worn). As discussed in Chapter 2, the bottom-up strategy typically leads to underreporting because of errors of omission. However, the top-down strategy usually leads to overreporting because of the presence of stability bias (Burton & Blair, 1991; Menon, 1994), even though respondents can occasionally engage in an "anchor and adjust" strategy when they use their present status as a benchmark against which they adjust past events based on their perception of stability or change (Ross, 1989).

Respondents sometimes rely primarily on the response categories to formulate an answer. When respondents are presented with a frequency scale to report on a past behavior, they use the scale as an additional source of information, assuming that the range of the scale reflects the population distribution of the behavior of interest and the middle point of the scale represents the average for the population. This strategy results in higher frequency estimates when a high-frequency scale is presented, and lower estimates when a low-frequency scale is presented. For example, in a study by Schwarz et al. (1985), participants were asked about their daily hours of TV watching and presented either with a low frequency (Up to 0.5 hours, 0.5–1 hours, 1–1.5 hours, 1.5–2 hours, 2–2.5 hours, and more than 2.5 hours) or high frequency (Up to 2.5 hours, 2.5–3 hours, 3–3.5 hours, 3.5–4 hours, 4–4.5 hours, more than 4.5 hours) alternatives. Responses were recoded to reflect estimates of 2.5 hours or more than 2.5 hours. As expected, the scale range had a significant effect on reporting – only 16.2% of the low-frequency

condition respondents reported more than 2.5 hours of TV watching, while that percentage more than doubled (37.5%) for the high-frequency condition. Interestingly however, in a follow-up study, [Chassein et al. \(1987\)](#) observed that the scale effect disappeared when respondents were given a chance to cue their memories by browsing through the previous week's TV program. Nonetheless, multiple studies have demonstrated that scale effects are robust and replicate over multiple domains (e.g., [Billet & Waterplas, 1988](#); [Schwarz, 1990](#); [Schwarz & Bienias, 1990](#)). Furthermore, proxy reports (reports about others' behaviors) are more affected by scale frequency than self-reports because people generally have better knowledge of their own behaviors than those of others ([Schwarz & Bienias, 1990](#)).

The choice of estimation strategy respondents would employ depends on multiple factors, such as number of events to be reported, similarity and regularity of events, length of the recall period, and time to answer. Any of the recall improvement strategies discussed in the previous section would help facilitate recall, regardless of the recall strategy.

Summary

In this chapter, we discuss reasons for recall error and strategies to improve retrieval and estimation strategies respondents engage in when answering behavioral questions. We alert the reader to the fact that response alternatives may add unintended information to the response processes and that open-ended questions might be preferable for behavioral reports where respondents are expected to provide numeric answers (thus, no extra coding is needed). Instead, some researchers may want to resort to using vague quantifiers, such as "often," "sometimes," and "most of the time"; however, vague quantifiers are not recommended for behavioral questions because they do not carry a unified meaning across respondents and often represent different frequencies in different contexts – for example, "frequently" is less frequent for rare events, such as a volcanic eruption, than frequent ones, such as thunderstorms.

As discussed, both the bottom-up and the top-down estimation strategies are imperfect and lead to different types of error, but the methods for improving recall often help with data accuracy. One variable under researchers' control is the length of the reference period; manipulating it based on what strategy respondents are more likely to engage in (enumeration for infrequent events and estimation for frequent ones) may improve recall quality by offering shorter reference periods for events likely to be estimated and longer for events likely to be counted (to avoid telescoping). Another variable under researchers' control is the amount of time given to respondents to answer a question in interviewer-administered surveys – results from multiple studies suggest that respondents who try to recall specific events retrieve more information when given more time. However, this strategy is unlikely to be very successful for respondents who engage in estimation rather than enumeration.

Exercises

- 1 Discuss event characteristics that can affect recall. Compare and contrast two strategies for improving recall in the context of event characteristics and describe in what situations you would prefer one over the other.
- 2 What are the main strategies respondents employ when answering behavioral frequency questions? When are they more likely to engage in one over the other?
- 3 What are the dangers of using behavior frequency scales?

4 Writing Attitudinal Questions

Introduction

What Is an Attitude?

In addition to facts and behaviors, survey researchers are also interested in measuring opinions and attitudes. Opinions and attitudes are likely based on facts, but facts are not directly referred to in attitudinal questions and likely are not consciously sought by respondents. Attitudes are defined as “an enduring positive or negative feeling about some person, object or issue” (Petty & Cacioppo, 1981, p. 7). Attitudes have direction and intensity and consist of three components – cognitive, affective, and action (an example is shown in the text box). The cognitive component refers to what the respondent knows about the attitude object. That knowledge may not be comprehensive, or even correct, and is limited to what the respondent has accessible in memory. Somewhat related to the cognitive component is the affective component – it is the primary dimension measured in surveys and reflects how respondents feel toward the attitude object. Finally, the action component reflects the respondent’s willingness or likelihood to behave in a way related to the attitude object. Unlike factual and behavioral questions discussed in [Chapter 3](#), attitudinal questions invoke subjective responses, so whether respondents provide a truthful response cannot be measured objectively.

Example: Attitudes toward abortion

Cognitive component: 10,000 abortions occur each year in the United States.

Affective component: Every woman should be able to get an abortion.

Action component: Intention/likelihood of getting an abortion.

Theories on Attitude Formation and Retrieval

How attitudes are reported by respondents has interested psychologists for a long time. Various theories are proposed to explain how people form and retrieve attitudes. Here we discuss two theories that are most relevant to the process of respondents answering attitudinal questions.

The File Drawer Model

One of the early theories on responses to attitude questions was the file drawer model (Wilson & Hodges, 1992). According to the model, evaluations and opinions on various topics are stored in the respondent's memory, and when needed, are looked up and retrieved, similar to extracting a file from a file drawer cabinet. When existing evaluations are not found in the file drawers, respondents rely on relevant information from the file drawers to form new judgments. The assumption this model makes is that most attitudes are preexisting and stable over time. However, these assumptions fail because findings from multiple surveys and experiments have demonstrated that attitudes change over time (e.g., Converse, 1964, 1970; Zaller, 1992), attitudes are extremely susceptible to context and question order effects (e.g., Sudman et al., 1996), and respondents express opinions about fictitious issues (Bishop et al., 1986; Schuman & Presser, 1981). Even though some respondents (particularly those with strong attitudes) answer attitudinal questions by retrieving an existing evaluation (as if from a file drawer), others generate their responses through some other process that does not involve recall of an existing judgment.

The Belief Sampling Model

Proposed by Tourangeau et al. (2000) and further elaborated in Tourangeau (2021), the model suggests that attitudes are a “haphazard assortment” (Tourangeau et al., 2000, p. 179) of existing evaluations, vague impressions, general values, and relevant feelings toward an attitude object. Thus, attitudes may be created through different paths, frequently dependent on the accessibility of information cued by preceding questions or chronically available, time respondents have on task, and respondents' motivation. As such, attitudes are inherently unstable and unreliable, as they tend to “overrepresent whatever considerations happen to be accessible when the question is asked” (Tourangeau et al., 2000, p. 181). Respondents with one-sided views on an issue are likely to produce more consistent attitudes because they will likely draw on the same considerations every time they are asked about their views. The attitudes of respondents with mixed underlying views, however, will be greatly impacted to the extent to which different considerations are brought to attention when asked about their views.

Strategies of Answering Attitudinal Questions

When respondents do not have an existing judgment to report, they might resort to other strategies for answering attitudinal questions as briefly described in Chapter 2: (1) general impressions/stereotypes or impression-based strategy (Sanbonmatsu & Fazio, 1990); (2) general values or top-down strategy (Zaller, 1992); or (3) specific beliefs and feelings that can be extrapolated to the attitude target or bottom-up strategy (Tourangeau & Rasinski, 1988).

Impression-based responses may occur when respondents do not have enough information or interest to form or retrieve an existing evaluation of the attitudinal target. In a study by Sanbonmatsu and Fazio (1990), respondents were presented with information about two department stores – one was described in a very favorable light, while the other was described in a very negative light – with the exception of their camera departments, which were described in the opposite light from the overall store: the terrible store had

an excellent camera department and vice versa. Respondents were asked to report from which store they would choose to buy a new camera. When they had to respond under pressure, most study participants selected the wonderful store with the subpar camera department; however, when given time to think and not rely on their general impressions, participants made choices consistent with what they knew about each camera department.

Based on the top-down retrieval strategy, general values come to play when respondents do not have a preexisting opinion toward a narrow attitude target but can express an opinion that reflects their general values and ideological norms. Examples of such attitudes include opinions on various political issues, which are highly susceptible to question wording – for example, in a famous experiment by [Rugg \(1941\)](#) replicated over time, respondents were randomly assigned to one of the following versions of the same question:

Do you think the United States should allow public speeches against democracy?

Do you think the United States should forbid public speeches against democracy?

Interestingly, 75% of respondents answered “no” to the first question, indicating that the United States should *not allow* speeches against democracy. By contrast, only 45% of respondents answered “yes” to the second question indicating that the United States should *forbid* speeches against democracy. Such seemingly small wording changes can alter how respondents apply their general values and norms on a narrow topic for which they might not have a preexisting opinion. In this particular instance, “forbid” is such a strong word that seems to contradict the principles of “democracy.” Respondents, not paying much attention to the specific attitude object, were influenced by this controversy to say no to the second question.

Similarly, respondents may construct an opinion based on an inference from a specific belief or beliefs they hold. Studies that explore this mechanism have looked at responses provided to open-ended questions and found that, for example, views on the local economy contained specific considerations of the local unemployment rate and job growth ([Mason et al., 1994](#)). Studies also examine reaction time (how long it took respondents to provide an answer) when respondents are primed with a related issue, an unrelated issue, different aspects of the same issue, or the same aspect of an issue. In a study by [Tourangeau et al. \(1991b\)](#) participants were asked to agree or disagree with two statements and recorded how long it took to provide a response to the second item. The study found increased response speed when the prime and target questions were very closely related, such as when representing the same aspect of an issue (e.g., abortion is murder and life begins at conception). The authors argue that speed of response would not have decreased if responses to both questions stem from a general evaluation rather than a specific belief and that respondents likely retrieve specific considerations for the speed of response to be affected by how related the two questions are.

Impact of Context on Answering Attitudinal Questions

Based on the belief sampling model, attitudes are unstable and dependent on information available in working memory. Context (in particular, question order) plays an important role at each stage of the survey response model. Context effects become an issue at the comprehension stage when the question is about an ambiguous or

unfamiliar attitude target (e.g., the study by Strack, Schwarz, & Wänke, 1991, described in [Chapter 2](#)). As working memory degrades with age, context effects are less pronounced in older respondents than in young ones (Knauper, 1998). On the flip side, chronically accessible information, such as previously formed judgments, will be less affected by the context.

At the retrieval stage, the information to form an attitude must be available in working memory triggered by current or prior questions. Because prior questions on related topics affect what is considered for subsequent questions, their impact is demonstrated by studies that focus on response time – answering a target question on a topic similar to the preceding questions makes respondents answer more quickly (e.g., [Bassili & Fletcher, 1991](#); [Judd et al., 1991](#); [Tourangeau et al., 1989a, 1989b](#)). Because quick response times are associated with decreased respondent burden, a best practice in questionnaire design is to group questions on a similar topic (thus, questions that use the same attitude object together).

Context effects in forming attitudes are likely the most pronounced at the judgment stage, when respondents need to make an evaluation relative to something else. The direction of the context effect on a response to a subsequent question may vary – we distinguish between **assimilation** and **contrast** effects. Assimilation effects occur when responses to the target question are in the same direction as responses to preceding (context) questions; conversely, contrast effects (including subtraction-based contrast effects as noted in [Tourangeau et al. \(1991a\)](#)) occur when responses to the target question move away from responses to the preceding questions.

An interesting example of how question order affects the norm invoked in making a judgment is a study originally published by [Hyman and Sheatsley in 1950](#) and replicated by many over the years (e.g., [Schuman & Ludwig, 1983](#); [Schuman & Presser, 1981](#)). Respondents were asked two questions, with varied order:

Do you think the United States should let communist reporters from other countries come in here and send back to their papers the news as they see it?

Followed, or proceeded by:

Do you think a communist country like Russia should let American newspaper reporters come in and send back to their papers the news as they see it?

[Figure 4.1](#) presents the percentage of respondents who answered “yes” to the communist reporters’ question by question order. The results show a dramatic shift in responses of 37% in the original study, depending on whether the communist reporters’ question is asked first or second. When the communist question is asked first, it evokes beliefs and attitudes at the time toward Russia and communism in general. However, when the same question follows the American reporters question, the norm of reciprocity is invoked, requiring the same treatment for communist reporters. This is an example of a *comparison-based assimilation effect*.

A study by [Schwarz et al. \(1990\)](#) is another example of context effects at the judgment stage. Study participants were asked to rate on a scale from 1 to 9 the typicality of wine, milk, and coffee as German beverages. These target questions were preceded by a context question asking typicality of beer or vodka, frequency of beer or vodka drinking in Germany, or caloric content of beer or vodka in a 2 (beer vs. vodka) $\times$ 3 (typicality rating, consumption frequency, caloric content) factorial design, where participants were randomly assigned

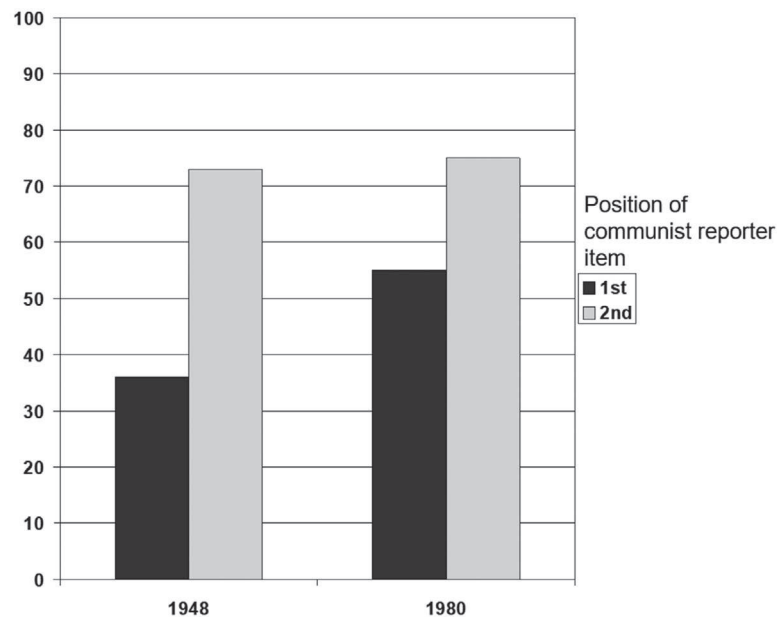


Figure 4.1 Percent Respondents Agreeing that Communist Reporters Should be Given Access, by Question Order and Study Year.

to a condition. Table 4.1 presents the results of the averaged typicality score for wine, milk, and coffee across conditions. The study demonstrates that respondents' ratings of typicality of wine, milk, and coffee were influenced by the extreme anchors (vodka or beer) used in the preceding context question especially when the context and target questions tapped into the same underlying dimension (typicality). Compared to vodka, which is not a typical German drink, wine, milk, and coffee were rated higher on the typicality dimension than when compared to beer, which is a very typical German drink. However, there was no contrast effect when the preceding context question asked about a different evaluative dimension (i.e., caloric content).

At the response mapping stage, context affects how respondents map their attitude on the provided response scale and whether they change their true answers beforehand. An example of how contextual information influences responses at the response

Table 4.1 Mean Typicality Rating Averaged Over Wine, Milk, and Coffee by Experimental Condition

Preceding questions	Context stimulus	
	Vodka	Beer
Typicality Rating	5.4	4.4
Consumption frequency	5.3	4.9
Caloric content	4.3	4.5

Note: Measured on a scale from 1 to 9, where 1 is not at all typical and 9 is very typical.

mapping stage is respondents' reluctance to provide negative evaluations, resulting in overuse of the positive end of a scale (positivity bias). A study by Schwarz et al. (1991a) presented respondents with two sets of rating scale, with different numeric end points – one set had values from –5 to +5, while the other ranged from 0 to 10. Despite having the same verbal endpoints on both scales (not at all successful and extremely successful), respondents largely utilized the positive end of the scale. When the negative numeric label was used for the “not at all successful” verbal label, it likely implied presence of failure rather than lack of success. Furthermore, when the numeric labels (e.g., ± 5) suggested a bipolar scale, while the verbal labels suggested a unipolar scale (e.g., not at all successful to extremely successful), respondents used only part of the scale, as supported by follow-up studies (e.g., Mazaheri & Theuns, 2009; O'Muircheartaigh et al., 1995).

Considerations for Developing Attitudinal Questions

Issues with attitudinal questions mostly occur at the judgment stage of the survey response process described in Chapter 2; however, questionnaire designers should watch out for problems that could occur at each of the four stages.

At the comprehension stage, attitude object **ambiguity** could be a major issue. In a study by Belson (1981) asking respondents to express an opinion on whether “children suffer any ill effects from watching programs with violence in them,” participants demonstrated very different interpretation of the terms “children,” “ill effects,” and “violence.” To avoid such inconsistencies across respondents, a best practice is to define the attitude target and then follow up with a question about respondents' opinions. A possible fix to the earlier example could be:

Do you think children under 14 are affected by watching videos or movies of people inflicting or sustaining physical injuries, using abusive behaviors or language, or damaging property?

However, question writers need to evaluate the tradeoffs between clarity and complexity and should strive for clearly specified concepts using simple language without making the question too long or too syntactically complicated.

As illustrated in the earlier section, the report of attitudes is very sensitive to context – if most attitudes are formed on the spot (in accordance with the belief sampling model), preceding questions will determine what information is currently active in the respondent's mind and what remains temporarily available in working memory.

To avoid unexpected context effects, we recommend writing simple questions, with a clear attitude target. Grouping questions by topic also helps control context effects in most modes. When there are concerns that the question order may influence the intended interpretation of the questions, a best practice is to randomize the order to estimate the effect.

A few other considerations, independent of the survey response process, should be taken into account when developing questions to measure attitudes and opinions.

Question balancing provides an opportunity for respondents to consider a counter-argument. In its simplest form, a balanced question contains the positive and negative responses (argument and counter-argument):

Do you feel a woman should be allowed to have an abortion in the early months of pregnancy if she wants one, or do you feel this should not be allowed?

Alternatively, the question may present a substantive counter-argument, such as *or do you feel this should depend on the reason for abortion?* One concern with substantive counter-arguments is that it is nearly impossible to perfectly balance the question and it may not even represent the respondent's position on the issue. Because the argument introduces information that otherwise may not have been available, it is included in the respondent's evaluation of the attitude object, so what alternative is chosen among sets of counter-arguments really matters. For example, in experiments by [Schuman and Presser \(1981\)](#), respondents were randomly assigned to one of the following questions:

If there is a serious fuel shortage this winter, do you think there should be a law requiring people to lower the heat in their homes, or do you think this should be left to individual families to decide?

If there is a serious fuel shortage this winter, do you think there should be a law requiring people to lower the heat in their homes, or do you oppose such a law because it would be too difficult to enforce?

The level of support for that law differed significantly between the two conditions – 35% of those who received the first version of the question favored the law, while only 25% of those who received the second version of the question endorsed it. Generally, question balancing is considered a best practice and is especially useful to minimize response acquiescence¹ – instead of presenting respondents with a single statement with which they need to agree or disagree, question balancing presents a forced choice. Furthermore, many studies have demonstrated a shift toward the counter-argument, when one is presented; for example, in the series of experiments noted earlier, [Schuman and Presser \(1981\)](#) randomly assigned respondents to unbalanced versus balanced questions and reported significant changes in endorsement rates toward the counter-argument, when it was offered.

Attitudinal questions often use response scales as response options. Response scales consist of a series of numeric or verbal values conveying an order. An important characteristic of scales as used in social sciences is **scale polarity** – the main distinction being between *bipolar and unipolar scales*.

Bipolar scales reflect two opposing alternatives and present a clear conceptual midpoint, collecting information on both the positive and negative endpoints. The intention is to measure the direction (side of the scale) and intensity (distance from the midpoint) of respondent's opinion on an issue. An example of a bipolar scale is shown below:

Very Positive ↔ Positive ↔ Mixed ↔ Negative ↔ Very Negative

In contrast, unipolar scales reflect gradation on the same dimension and do not present a clear conceptual midpoint. As a result, such scales often seem unbalanced:

Very Positive ↔ Positive ↔ Somewhat Positive ↔ Not at all Positive

Unipolar scales are easier to construct, but do not provide an idea to which of the two opposing ends the respondent belongs. Thus, two separate questions for the positive and negative directions need to be asked.

Yet another approach for asking questions utilizing a bipolar scale is **branching** (or **scale unfolding**) where instead of presenting the full scale from the beginning, respondents are first asked about direction to determine whether they agree or disagree with a statement, and then about intensity of their agreement or disagreement (e.g., completely, generally agree, or disagree). Research on the effects of branching over data quality is mixed – some present evidence that branching is beneficial (e.g., Krosnick & Berent, 1993), while others show no effect or negative impact on validity (Gilbert, 2015; Wang & Krosnick, 2019). Some authors recommend that when branching is used, the middle category should also be presented (Malhotra et al., 2009). We recommend careful consideration when to use bipolar instead of unipolar scales, alerting the reader that branching would also increase the number of items (similar to asking two unipolar questions) and may have an impact on overall cost (Gilbert, 2015).

The decision on what kind of scale to use is often practical and dependent on the attitude object that is measured. In surveys, the traditional belief is that using two unipolar scales to measure the positive and negative directions is better than one bipolar scale (Alwin et al., 2018; Cacioppo et al., 1997; Gannon & Ostrom, 1996; Mazeheri & Theuns, 2009; Schaeffer & Thompson, 1992; Solomon, 1978). From a practical perspective, clear antonyms that are used in bipolar scales do not often exist, and the use of negative adjectives (e.g., exceptional-unexceptional) is not equivalent to an antonym comparison (e.g., exceptional-ordinary) (Paradis et al., 2009). Thus, using unipolar scales provides more flexibility to measure the positive and negative dimensions of a phenomenon and yields more reliable results (Alwin et al., 2018).

There is significant research on the use of **verbal labels** in addition to the numeric ones of a response scale. There is general support for the finding that verbal labels increase scale reliability (e.g., Alwin, 2007; Lau, 2018; Menold et al., 2014) and validity (e.g., Lau, 2018), as they unify the meaning of the scale points across respondents. The number of scale points also dictates the extent to which verbal labels can be used for all, or only the end categories. Based on studies of response order effects and scale reliability, the recommended number of scale points for both bipolar and unipolar scales is five (e.g., Pasek & Krosnick, 2010; Revilla et al., 2014), with at least the end and middle categories verbally labeled.

The inclusion of a **middle category** in the response scale depends on the type of rating scale. As previously noted, bipolar scales have a meaningful midpoint that should be included. Labels for the middle category vary from “neutral,” “neither-nor,” and “middle of the road” to “both equally,” but they all mark a transition from positive to negative (or vice versa). Often, the middle category is used as the equivalent of “Don’t Know” or “Undecided” (e.g., Dykema et al., 2019; Sturgis et al., 2014) and similar to “Don’t Know,” many studies have reported endorsement of the middle category when offered (e.g., Converse & Presser, 1986; Kalton & Schuman, 1982). Schaeffer and Thomson (1992) suggest that the middle category of a bipolar scale can mean many things – from a respondent’s indifference to ambivalence; however, not including a middle category does not improve data quality (Wang & Krosnick, 2019).

In series of experiments conducted at the University of Michigan, Schuman and Presser (1981) examine whether the location of the middle category (in the middle of the response scale or at the very end) has an impact on responses, whether the number of substantive response options affects the endorsement of the middle category, and whether the middle category has an effect on the rest of the response options in interviewer-administered surveys. The results suggest that the location of the middle category does

Table 4.2 Example Full and Quasi Filters

<i>Full filter</i>	<i>Quasi filter</i>
Here is a statement about another country. Not everyone has an opinion on this. If you do not have an opinion, just say so. Here is the statement: The Russian leaders are trying to get along with America. Do you have an opinion on that?	Here is a statement about another country: The Russian leaders are trying to get along with America. Do you agree, disagree or not have an opinion?

make a difference – it is endorsed significantly more when presented at the end of the question, likely because of recency effects. Furthermore, the authors report that the more points there are on a scale, the less likely the middle category is to be selected (based on significant differences in the endorsement of the middle category in a 3-point vs. 5-point scale), and that the middle category gets selected when offered, but its presence does not seem to have an effect on the substantive responses.

One way to avoid over-endorsement of the middle category is to include knowledge questions that can filter out respondents who do not have an opinion on the issue. We distinguish between **full filters and quasi filters**. Full filters typically present a statement to the respondent and ask whether they have an opinion on that. Only those who have an opinion proceed to the question that measures that opinion. Table 4.2 presents examples of full and quasi filters from the Schuman and Presser (1981) experiments. Quasi filters present the “no opinion” option within the question stem, along with the opinion options. Experiments on using full versus quasi filters demonstrate that full filters tend to filter out significantly more respondents (e.g., Schuman & Presser, 1981). Hippler and Schwarz (1989) report that strongly worded filters, such as “Have you already heard or read enough about it to have an opinion?” might suggest to respondents that a great deal of knowledge is needed to answer the follow-up attitudinal questions in comparison to a simple filter such as “Do you have an opinion on this or not?” and thus screen out more respondents.

Acquiescence, or agreeing response bias, is a respondent’s tendency to agree or answer affirmatively regardless of the presented attitudinal statements. Acquiescence occurs regardless of mode of data collection (e.g., Liu & Keusch, 2017), or number of response categories (e.g., Schuman & Presser, 1981), and is mostly associated with Likert-type of scales (e.g., strongly agree-strongly disagree). Acquiescence may be dependent on the order of the response options; for example, Liu and Keusch (2017) report that acquiescence is more prevalent in web mode than face-to-face mode when the “agree” end of the scale is presented first. However, there are several ways to detect and minimize acquiescence. One is to use **reverse coding** – for example, respondents may be asked to agree or disagree with two contradictory statements:

Teenagers should be allowed to stay out at night as long as they want to.

Several questions later, respondents may be presented with the exact opposite statement:

Teenagers should have curfew.

Those acquiescing will agree with both contradictory statements and their responses may be flagged to later decide whether to exclude from analyses or correct for measurement error.

Another approach to avoid acquiescence is to move away from Likert scales and use **construct-specific** questions rather than agree-disagree statements:

Would you say that most men are better suited emotionally for politics than are most women, that men and women are equally suited, or that women are better suited than men in this area?

Rather than:

Tell me if you agree or disagree with this statement: Most men are better suited emotionally for politics than are most women.

(General Social Survey, 2002)

Overall, recent research consistently recommends the use of construct-specific questions because of their higher reliability and validity in comparison to agree-disagree statements (e.g., [Revilla & Ochoa, 2015](#); [Saris et al., 2010](#)). Even though there are several studies that fail to show differences in reliability or validity (e.g., [Lelkes & Weiss, 2015](#)), we still recommend construct-specific questions for attitudes.

Summary

Writing attitudinal questions presents multiple challenges, because attitudes are often created on the spot and the attitude target is vague, making responses highly dependent on context and what is activated in a respondent's working memory. To avoid unwanted context effects and other issues such as acquiescence, we recommend specifying the attitude target as clearly as possible and asking item-specific questions rather than agree-disagree questions. Furthermore, if there are concerns about question groupings within a content topic, we recommend randomizing the order where an impact is expected. Finally, using verbal labels in response scales tends to increase data quality and consistency across modes and we recommend it whenever possible.

Exercises

- 1 What are the advantage of the belief sampling model over the file drawer model?
- 2 Discuss comparison-based contrast effects and assimilation effects, providing examples from your daily life.
- 3 Balance the following question, then create two counter-arguments:

Do you think it is the responsibility of the federal government to make sure all Americans have health care coverage?

(Pew Research Center, <https://www.pewresearch.org/2024/06/24/americans-views-of-governments-role-persistent-divisions-and-areas-of-agreement/>)

Note

- 1 Respondent's tendency to say "yes" or "agree" to attitudinal questions.

5 Writing Demographic and Knowledge Questions

Introduction

Questions that measure demographic characteristics are often considered quasi-factual questions, because they vary in terms of objective and subjective elements. Examples of characteristics that have objective elements are age, sex at birth, income, and education status; examples of characteristics that contain subjective elements include identification of one's race and ethnicity and gender identity. Demographic questions in surveys are collected for three main purposes: (1) to screen out respondents who do not belong to the target population (e.g., the National Survey of Family Growth seeks 15–45-year-old respondents), (2) to make weighting adjustments to known population totals, and (3) to compare subgroups on survey substantive outcomes. Those goals, along with mode of data collection, drive the placement of demographic questions. Typically, demographic questions are presented at the very end, because they are not very engaging and some are perceived as sensitive or intrusive. However, such questions can be placed at the beginning of a survey when needed to determine respondent eligibility, selection of respondents, and the questionnaire path a respondent should take (e.g., whether respondents are administered a male- or female-specific questionnaire in the National Survey of Family Growth is determined by their response to the sex at birth screener question).

Knowledge questions are factual in nature, because they contain an objective truth that is known to the researcher. Uses of knowledge questions in surveys include screening out respondents who may not be knowledgeable enough on a topic to have an opinion, measuring political knowledge (e.g., the American National Election Study: <https://electionstudies.org/>), scientific knowledge (e.g., the General Social Survey: <https://www.norc.berkeley.edu/projects/gss.html>) literacy, or education level (e.g., the National Assessment of Education Progress: <https://nces.ed.gov/nationsreportcard/>), or obtaining information in the qualitative stage of survey development. This chapter introduces best practices for measuring selected demographic questions (race and ethnicity, gender identity, and income) and presents a discussion on uses and considerations for knowledge questions.

Demographic Questions

A movement to standardize demographic questions across US surveys has been attempted since 1975 based on the Social Science Research Council recommendation (Social Science Research Council, 1975). Using the same demographic questions across surveys not only allows for comparability but also enables survey practitioners to use data from federal surveys, such as the American Community Survey, in their weighting adjustments. Our discussion draws heavily from US Decennial Censuses and Office of Management and Budget (OMB) guidelines for certain demographic questions and may not have direct relevance for surveys outside of the United States.

DOI: [10.4324/9781003367826-6](https://doi.org/10.4324/9781003367826-6)

Measuring Race and Ethnicity

The meaning of race varies across countries. In the United States, race has been defined biologically, anthropologically, genetically, and, lately, socially. The US Census first introduced a race question in the 1790 Census, where it collected race data in three categories: free white people, all other free persons (including free Black and a subset of Native Americans under US jurisdiction), and slaves. The free people categories in the 1850 census included white, black, or mulatto and introduced a separate form for enslaved people, who were counted under their owners. The first post-Civil War census, the 1870 census, for the first time did not count slaves and introduced “Chinese” to count people of Asian identity. In fact, each decennial census has presented a slightly different way of collecting race and ethnicity, reflecting societal changes. The 1997 OMB standards on race and ethnicity signaled a paradigm shift to self-identification, employing the social definition of race/ethnic belonging and allowing respondents to select more than one racial category. Ethnicity is

Person 1

5. Please provide information for each person living here. If there is someone living here who pays the rent or owns this residence, start by listing him or her as Person 1. If the owner or the person who pays the rent does not live here, start by listing any adult living here as Person 1.

What is Person 1's name? *Print name below.*

First Name MI

Last Name(s)

6. What is Person 1's sex? Mark ☒ ONE box.

☐ Male ☐ Female

7. What is Person 1's age and what is Person 1's date of birth? For babies less than 1 year old, do not write the age in months. Write 0 as the age.

Age on April 1, 2020 years months days Year of birth

Print numbers in boxes.

Month Day Year

→ NOTE: Please answer BOTH Question 8 about Hispanic origin and Question 9 about race. For this census, Hispanic origins are not races.

8. Is Person 1 of Hispanic, Latino, or Spanish origin?

☐ No, not of Hispanic, Latino, or Spanish origin

☐ Yes, Mexican, Mexican Am., Chicano

☐ Yes, Puerto Rican

☐ Yes, Cuban

☐ Yes, another Hispanic, Latino, or Spanish origin – *Print, for example, Salvadoran, Dominican, Colombian, Guatemalan, Spaniard, Ecuadorian, etc.*

9. What is Person 1's race? Mark ☒ one or more boxes AND print origins.

☐ White – *Print, for example, German, Irish, English, Italian, Lebanese, Egyptian, etc.*

☐ Black or African Am. – *Print, for example, African American, Jamaican, Haitian, Nigerian, Ethiopian, Somali, etc.*

☐ American Indian or Alaska Native – *Print name of enrolled or principal tribe(s), for example, Navajo Nation, Blackfeet Tribe, Mayan, Aztec, Native Village of Barrow, Inupiat Traditional Government, Nome Eskimo Community, etc.*

☐ Chinese ☒ Vietnamese ☐ Native Hawaiian

☐ Filipino ☐ Korean ☐ Samoan

☐ Asian Indian ☐ Japanese ☐ Chamorro

☐ Other Asian – *Print, for example, Pakistani, Cambodian, Hmong, etc.*

☐ Other Pacific Islander – *Print, for example, Tongan, Fijian, Marshallese, etc.*

☐ Some other race – *Print race or origin.*

Figure 5.1 Example Race and Ethnicity Questions from the United States 2020 Census.

defined by OMB as either “Hispanic or Latino” or “Not Hispanic or Latino.” Hispanic or Latino is defined as Cuban, Mexican, Puerto Rican, South or Central American, or other Spanish culture or origin, regardless of race.

The 2020 Census race and ethnicity questions and categories are presented in questions 8 and 9 in [Figure 5.1](#).

The 2024 revised OMB guidelines ([spd15revision.gov](https://www.spd15revision.gov)) recommend one combined question for race and ethnicity, allowing respondents to select as many response categories as they want. The question can be presented in three formats, as shown in [Figure 5.2](#).

1. Minimum Categories, Multiple Detailed Checkboxes and Write-In Response Areas with Example Groups

What is your race and/or ethnicity?
Select all that apply and enter additional details in the spaces below.

☐ **American Indian or Alaska Native** – *Enter, for example, Navajo Nation, Blackfeet Tribe of the Blackfeet Indian Reservation of Montana, Native Village of Barrow Inupiat Traditional Government, Nome Eskimo Community, Aztec, Maya, etc.*

☐ **Asian** – *Provide details below.*

☐ Chinese

☐ Asian Indian

☐ Filipino

☐ Vietnamese

☐ Korean

☐ Japanese

Enter, for example, Pakistani, Hmong, Afghan, etc.

☐ **Black or African American** – *Provide details below.*

☐ African American

☐ Jamaican

☐ Haitian

☐ Nigerian

☐ Ethiopian

☐ Somali

Enter, for example, Trinidadian and Tobagonian, Ghanaian, Congolese, etc.

☐ **Hispanic or Latino** – *Provide details below.*

☐ Mexican

☐ Puerto Rican

☐ Salvadoran

☐ Cuban

☐ Dominican

☐ Guatemalan

Enter, for example, Colombian, Honduran, Spaniard, etc.

☐ **Middle Eastern or North African** – *Provide details below.*

☐ Lebanese

☐ Iranian

☐ Egyptian

☐ Syrian

☐ Iraqi

☐ Israeli

Enter, for example, Moroccan, Yemeni, Kurdish, etc.

☐ **Native Hawaiian or Pacific Islander** – *Provide details below.*

☐ Native Hawaiian

☐ Samoan

☐ Chamorro

☐ Tongan

☐ Fijian

☐ Marshallese

Enter, for example, Chuukese, Palauan, Tahitian, etc.

☐ **White** – *Provide details below.*

☐ English

☐ German

☐ Irish

☐ Italian

☐ Polish

☐ Scottish

Enter, for example, French, Swedish, Norwegian, etc.

Figure 5.2 Race/Ethnicity Question Stipulated in the 2024 OMB Guideline. (Continued)

2. Minimum Categories Only and Examples

What is your race and/or ethnicity?
Select all that apply.

☐ **American Indian or Alaska Native**
For example, Navajo Nation, Blackfeet Tribe of the Blackfeet Indian Reservation of Montana, Native Village of Barrow Inupiat Traditional Government, Nome Eskimo Community, Aztec, Maya, etc.

☐ **Asian**
For example, Chinese, Asian Indian, Filipino, Vietnamese, Korean, Japanese, etc.

☐ **Black or African American**
For example, African American, Jamaican, Haitian, Nigerian, Ethiopian, Somali, etc.

☐ **Hispanic or Latino**
For example, Mexican, Puerto Rican, Salvadoran, Cuban, Dominican, Guatemalan, etc.

☐ **Middle Eastern or North African**
For example, Lebanese, Iranian, Egyptian, Syrian, Iraqi, Israeli, etc.

☐ **Native Hawaiian or Pacific Islander**
For example, Native Hawaiian, Samoan, Chamorro, Tongan, Fijian, Marshallese, etc.

☐ **White**
For example, English, German, Irish, Italian, Polish, Scottish, etc.

3. Minimum Categories Only

What is your race and/or ethnicity?
Select all that apply.

☐ **American Indian or Alaska Native**

☐ **Asian**

☐ **Black or African American**

☐ **Hispanic or Latino**

☐ **Middle Eastern or North African**

☐ **Native Hawaiian or Pacific Islander**

☐ **White**

Figure 5.2 (Continued)

Measuring Sex and Gender Identity

Similar to race, how we collect data on a respondent's sex and gender has evolved over time, from a simple binary category (male-female) question, based on the assumptions that sex and gender do not differ from one another and sex/gender is objective in nature and does not change over time, to series of questions measuring sex assigned at birth, gender identity, and sexual orientation, based on the latest research that defines sex as a biological trait and gender as social and behavioral characteristics (e.g., [Hall et al., 2021](#); [Schudson et al., 2019](#)).

Sexual orientation is related to both sex and gender because it is determined based on relationship of one's sex or gender and that of their partner. The National Academies of Science recommends the following measures (<https://nap.nationalacademies.org/catalog/26424/measuring-sex-gender-identity-and-sexual-orientation>):

Sex at Birth and Gender Identity

Q1: What sex were you assigned at birth, on your original birth certificate?

- *Female*
- *Male*

(Don't know)

(Prefer not to answer)

Q2: What is your current gender? [Mark only one]

- *Female*
- *Male*
- *Transgender*
- *[If respondent is AIAN:] Two-Spirit*
- *I use a different term: [free text]*

(Don't know)

(Prefer not to answer)

Sexual Orientation

Which of the following best represents how you think of yourself? [Select ONE]:

- *Lesbian or gay*
- *Straight, that is, not gay or lesbian*
- *Bisexual*
- *[If respondent is AIAN:] Two-Spirit*
- *I use a different term [free-text]*

(Don't know)

(Prefer not to answer)

Measuring Income

Household income is an important concept that is highly correlated with a variety of economic, sociological, and health outcomes. However, it presents two major challenges – there are many potential sources of income, and respondents do not like to disclose how much money they have. As a result, questions measuring income are associated with high levels of missing data. In an attempt to combat these challenges and minimize measurement error, some federal surveys employ the unfolding brackets approach to collect income data. In this approach, if the respondent does not answer the initial income question, a follow-up question presenting a rounded figure is asked and respondents have to indicate if their household income is above or below that number. Similar questions could follow until a closer range for the household income can be determined. This method has been found effective in reducing nonresponse and uncertainty around income measures in many surveys (e.g., Juster & Smith, 1997; Pleis et al., 2006; Wang, 2010). Figure 5.3 presents an example from the Health and Retirement Survey.

An alternative approach many surveys employ is to provide income ranges as response options from the very beginning, instead of asking an open-ended question. For example, Figure 5.4 presents the household income question from the National Crime Victimization Survey.

Q015 What was your income from self-employment, before taxes and other deductions, in LAST CALENDAR YEAR? Please include income you actually received from the business and any profits that may have been left in the business. [INSTR: DO NOT PROBE DK/RF.] \$_____.00 GO TO Q019 DK RF
Q016-Q018 Question text: (Thinking about your self-employment income in [LAST CALENDAR YEAR]): Did it amount to less than \$____, more than \$____, or what? PROCEDURES: 3Up, 2Up1Down, 1Up2Down BREAKPOINTS: \$5,000, \$10,000, \$25,000, \$100,000 RANDOM ENTRY POINT ASSIGNMENT [1 (\$5,000)] or [2 (\$10,000)] or [{NOT 1 and NOT 2} (\$25,000)] AT X041

Figure 5.3 Unfolding Brackets Example from the 2022 Health and Retirement Survey.

90. HOUSEHOLDINCOME (Asked of Household Respondent Only)

(Asked of household respondent.

Asked at 1st, 3rd, 5th, and 7th interview, or if never asked before. Asked at subsequent interviews if no or Don't Know/Refused at prior interview.)

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Which category represents the TOTAL combined income of all members of this HOUSEHOLD during the past 12 months? This includes money from jobs, net income from business, farm or rent, pensions, dividends, interest, Social Security payments, and any other money income received by members of this HOUSEHOLD who are 14 years of age or older.

1	☐ Less than \$5,000
2	☐ \$5,000 to \$7,499
3	☐ \$7,500 to \$9,999
4	☐ \$10,000 to \$12,499
5	☐ \$12,500 to \$14,999
6	☐ \$15,000 to \$17,499
7	☐ \$17,500 to \$19,999
8	☐ \$20,000 to \$24,999
9	☐ \$25,000 to \$29,999
10	☐ \$30,000 to \$34,999
11	☐ \$35,000 to \$39,999
12	☐ \$40,000 to \$49,999
13	☐ \$50,000 to \$74,999
14	☐ \$75,000 to \$99,999
15	☐ \$100,000 to \$149,999
16	☐ \$150,000 to \$199,999
17	☐ \$200,000 or more

Figure 5.4 Household Income Question from the 2022 National Crime Victimization Survey.**Knowledge Questions**

Using knowledge questions to screen out respondents who may not be knowledgeable enough to express an opinion on a topic is common in surveys. As such, knowledge screening questions can be placed anywhere in the questionnaire, or at the beginning of a series of attitudinal questions. The nature of such “no opinion” filters to a large extent determines how many respondents will get screened out. [Chapter 4](#) distinguishes between full filters and quasi-filters with examples of each presented in [Table 4.2](#).

When the intent is to measure political knowledge, the mode of data collection becomes an issue; in self-administered surveys, respondents can look up the correct answer or consult with others. Many studies have examined ways to prevent respondents from searching for the correct answer by specifically instructing them not to look up answers (e.g., [Motta et al., 2017](#); [Smith et al., 2020](#)), asking for explicit commitment not to look up responses (e.g., [Gummer et al., 2023b](#)), or using pictures of politicians, instead of names (e.g., [Munzert & Selb, 2017](#)). Although some studies evaluate the efficacy of the intervention based on self-reports, many are using paradata to detect new browser widows and tab switching (e.g., [Gummer & Kunz, 2022](#); also see [Chapter 7](#) for a discussion), but are of course unable to detect searches on another device. The current recommendation for minimizing answer look-up is using commitment statements such as:

It is important to us that you do not use any additional aids, such as the internet, to answer these questions. Do you agree to answer the next questions without additional aid?
(modified from [Gummer et al., 2023b](#))

Knowledge questions may be perceived as threatening by respondents because responses may present them in a negative light. To reduce this perceived threat, knowledge questions often begin with “Do you happen to know,” “As far as you know,” or “Can you recall, offhand” phrases. Such questions inherently suggest that not knowing the answer

is acceptable and explicit Don't Know responses are often included as response options to reduce the possibility of guessing in closed questions.

Questions that collect numeric information are best presented as open-ended questions to avoid the risk of respondents guessing correctly somewhere in the middle of the numeric scale. In the rare occasion where Yes/No questions are appropriate, we recommend asking several questions on the same topic to reduce the likelihood of successful guessing (Sudman & Bradburn, 1982).

Summary

Demographic questions vary in the extent to which subjective or objective elements are present, and this evolves over time, driven by societal changes. OMB has guidance on measuring important demographic characteristics, and it is a good practice to follow those recommendations, especially if federal survey data are used for weighting adjustments or comparisons. Knowledge questions have useful applications in surveys – screening out respondents, measuring knowledge of a particular field (e.g., political knowledge), and informing a qualitative portion of the questionnaire design process. However, the nature and wording of knowledge questions determine how many respondents will get screened out and should be considered carefully.

Exercises

- 1 Practice bracket unfolding by developing a question asking respondents how old they were when they first had sex.
- 2 Discuss possible uses of knowledge questions in surveys and considerations related to placement and mode of data collection.

6 Writing Sensitive Questions and Questions for Sensitive Populations

Introduction

It is common for general population surveys to include sensitive questions. For instance, the National Survey on Drug Use and Health (NSDUH) includes questions on health, drug use, and criminal justice involvement. The National Survey of Family Growth (NSFG) asks detailed questions on sexual behaviors, drug use, and abortion.

Tourangeau et al. (2000) define three types of sensitive questions. The first type concerns the intrusiveness of the content of the question. A survey item is intrusive when it invades a respondent's privacy and is considered inappropriate for normal conversations (see also [Tourangeau & Yan, 2007](#); [Yan, 2021](#)). For example, survey items about income, religion, and number of sexual partners may fall into this category. This type of sensitive question risks offending all respondents, regardless of their true status or their answers. The second type of sensitive question is related to the extent that disclosing the answer to a third party will put the person at risk of harm (Tourangeau & Yan, 2007; Tourangeau et al., 2000; [Yan, 2021](#)). For instance, people in the United States may not be comfortable providing their Social Security number because of potential stolen identity threats from disclosing that piece of information. Other disclosure risk-related examples include legal concerns about disclosing illegal drug use or citizenship status. The third type of sensitive question involves the concept of social desirability ([Tourangeau & Yan, 2007](#); Tourangeau et al., 2000; [Yan, 2021](#)). Social desirability is the extent to which survey questions cover topics conforming to accepted social norms ([DeMaio 1984](#); [Holtgraves, 2004](#)). Answers conforming to norms are considered socially desirable, whereas answers deviating from norms are socially undesirable. For example, society norms prescribe that good citizens are expected to vote and to *not* drink and drive. As a result, both questions on voter turnout and on driving under the influence of alcohol invoke social desirability concerns. Respondents are found to overreport socially desirable attitudes or behaviors (e.g., voting, church attendance, seat belt use, owning a library card, volunteering, and donating) and underreport socially undesirable attitudes or behaviors (e.g., drunk driving, abortion, and discrimination).

Sometimes questions are sensitive for more than one reason. For instance, questions about criminal justice involvement are sensitive because of the intrusiveness of the topic, negative consequences of disclosure, and social stigma associated with the topic ([Yan & Cantor, 2019](#)).

Question sensitivity is not a stable characteristic of survey questions (Yan, 2021). The same survey questions may be perceived as more or less sensitive by people from different countries and cultures (Andreenkova & Javeline, 2019; Johnson & van de Vijver, 2002; Triandis et al., 1965), by how they are asked (Kreuter et al., 2008a), and by who answers them (e.g., Kreuter et al., 2008a). In many cases, the perceived sensitivity of a survey item depends on whether the respondent has experienced the undesirable behavior of interest. Furthermore, some questions become more sensitive over time. For instance, questions on citizenship were included in the decennial Census form from 1890 to 2000, but it became such a sensitive issue that the US Supreme Court blocked the 2020 Census from asking it (Yan, 2021).

Several ways exist to measure sensitivity of a survey question. When data are available from other surveys asking about the same concept or from previous rounds of data collection, item nonresponse rates can be used to assess question sensitivity. This is because sensitive questions such as those about income (Yan et al., 2010), citizenship (Brown et al., 2019), voting intentions (Krumpal, 2013), and number of sexual partners (Tourangeau & Yan, 2007) are shown to produce a higher level of missing data than nonsensitive questions such as those about age and education (Brown et al., 2019; Krumpal, 2013; Tourangeau & Yan, 2007). Furthermore, results from validation studies comparing survey reports to external data are also used as evidence of the presence (or absence) of question sensitivity. For instance, validation studies found that survey respondents underreported abortions (e.g., Jones & Kost, 2007) and overreported voter registration and voting turnout (e.g., McDonald et al., 2017), confirming that questions on abortion and voter turnout are sensitive. Question writers also measure question sensitivity by asking coders or judges to determine how sensitive a question concept is (Tourangeau et al., 2000). People in the target population are sometimes asked directly how sensitive a survey question is. One way to accomplish this is to probe on perceived sensitivity of target survey questions during cognitive interviews (see Willis, 2005). Alternatively, debriefing questions asking actual respondents about their perception of the sensitivity of survey questions can be included at the end of the survey (e.g., Kreuter et al., 2008a).

Question sensitivity differs by culture, region, and country, and over time. Question sensitivity can be assessed through item nonresponse rates, validation studies, and reported perception of sensitivity.

Strategies for Asking Sensitive Questions

Yan (2021) described the impact of asking sensitive questions from the Total Survey Error perspective. Of particular importance to questionnaire design are the established empirical findings that people tend to *not* answer some sensitive questions such as those about income (Yan et al., 2010) and misreport their answers to other sensitive questions (Bradburn et al., 2004; Tourangeau & Yan, 2007; Tourangeau et al., 2000). As a result, when writing sensitive questions, the goals are to alleviate item nonresponse and to reduce socially desirable responding. Survey researchers have been developing methods and strategies to achieve these dual goals by changing either the environment in which sensitive questions are asked or the way sensitive questions are phrased. We group the strategies and techniques into three categories: (1) strategies related to survey design protocol, (2) strategies on preparing respondents for the task, and (3) strategies on writing sensitive questions.

The first two categories aim to change the environment in which respondents are asked sensitive questions, whereas the third category focuses on how to ask sensitive questions.

Strategies Related to Survey Design Protocol

This section discusses specific survey design features that can be used to change the environment in which sensitive questions are asked to reduce social desirability concerns and improve perceived privacy.

Survey design features shown to improve answers to sensitive questions include using self-administration, using a diary, and enhancing privacy and ensuring anonymity.

Using a Self-administered Mode

The survey literature has shown consistently that using a self-administered survey mode (e.g., web) increases disclosure of sensitive information and reduces socially desirable responding when compared to an interviewer-administered mode of data collection such as face-to-face interviewing (Bradburn et al., 2004; Tourangeau & Yan, 2007; Yan, 2021; Yan & Cantor, 2019). Self-administration removes interviewers from the survey-answering process, which eliminates respondents' need to look good in front of the interviewer and their concerns about privacy, and, thus, reduces social desirability bias in the resultant answers. For instance, data using NSFG have shown that women reported more abortions and miscarriages during the audio computer-assisted self-interviewing (ACASI) portion of the interview than when interviewed by an interviewer during computer-assisted personal interviewing (e.g., Yan & Tourangeau, 2022). Many large-scale surveys such as NSFG, NSDUH, and the Population Assessment of Tobacco and Health (PATH) include ACASI as part of the survey design for sensitive questions. One word of caution, however. Studies have shown that interviewers still affect answers to ACASI questionnaire even though interviewers only stay around and do not directly administer the ACASI questionnaire (e.g., West & Peytcheva, 2014).

In terms of different forms of self-administration, Tourangeau and Yan (2007) found no difference between computerized self-administration (e.g., web) and paper-and-pencil self-administration (e.g., mail surveys) in disclosure of sensitive information.

Collecting Data in Private and Ensuring Anonymity

The presence of other people (also called bystanders) at an interviewer-administered survey setting is found to increase misreporting to sensitive questions (e.g., Diop et al., 2015). In particular, the presence of parents during an interviewer-administered survey significantly increases social desirability bias in children's answers to sensitive questions (Tourangeau & Yan, 2007). As a result, Tourangeau and Yan (2007) recommend that survey interviews be conducted in a private setting, so that respondents are not concerned about someone else overhearing their answers to sensitive questions.

For self-administered surveys, respondents are usually encouraged to complete the self-administered survey in a private setting. Research has shown that *not* asking respondents to provide their full names (Ong & Weiss, 2000) or *not* showing any identifying

information (Yang & Yu, 2011) increases socially undesirable answers. We recommend taking additional steps to ensure anonymity for self-administered surveys.

Using a Diary

Bradburn et al. (2004) suggest using the diary approach to collect sensitive information because it does not invoke the same level of social desirability concerns as survey questions directly asking about sensitive topics. For instance, asking respondents to keep a diary on their time use yields more accurate information on their church attendance than a direct question asking them to self-report church attendance. Similarly, asking respondents to keep a diary on expenditures produces more accurate information on alcohol expenditure and drinking than direct survey questions. However, the major limitation with using this approach to reduce social desirability bias is that not all sensitive behaviors can be easily measured with such an approach. Besides, the diary approach has other undesirable measurement properties (see Yan & Machado, 2023, for a review of different types of diaries). For instance, diaries are not filled out every day as expected and desired by survey researchers, and the process of filling out a diary could create demand characteristic by changing the behavior of interest.

Strategies on Preparing Respondents for the Task

Commitment, Instructions, and Honesty Pledge

Commitment and instructions are two techniques proposed in the 1980s to improve respondents' question-answering performance in interviewer-administered surveys (Cannell et al., 1981). The idea behind instructions is to let respondents know what is expected from them and how they achieve those expectations. With commitment,

respondents are asked to formally commit to be diligent in doing what is expected from them. The two complement each other. Cannell et al. (1981) showed that combining commitment and instructions increased disclosure of embarrassing information for respondents regardless of their education status.

Two studies investigated the use of commitment and instructions on web surveys. Clifford and Jerit (2015) presented web respondents with a question:

"It is important to us that participants in our survey pay close attention to the materials. Are you willing to carefully read the materials and answer all of the questions to the best of your ability?"

and respondents could select Yes or No. They found that asking for commitment reduced the number of socially desirable responses only for respondents with low education.

Cannell et al. (1981) used instructions for two questions:

- *Let me just mention that, to be most accurate, you may need to take your time to think carefully before you answer.*
- *On this next question, we'd like to get numbers as exact as possible.*

Cibelli (2017) did not find an effect of commitment on improving disclosure of sensitive information in her web studies. It is important to note that these studies vary in how instructions were conveyed and how commitment was sought.

McDonald et al. (2017) tested a slightly different type of commitment – they asked a random half of their web respondents during a pre-election survey to take an honesty pledge (“Are you willing to keep track of whether you voted in the upcoming November election and to be honest when you report this?”). In the subsequent post-election survey, respondents were asked whether they voted in the November election. Comparing self-reports of voter turnout to records shows that the honesty pledge reduced the overreport of voter turnout by 5.3 percentage points for the full sample and 10.9 percentage points for nonvoters.

These studies demonstrate the potential of using commitment, instruction, and honesty pledges to reduce socially desirable responding. However, it is critical to test the instructions and procedures of obtaining commitment and pledge before they are fielded.

Cannell et al. (1981) described one commitment procedure used in a face-to-face survey:

That’s the last of this set of questions. The rest of the questions are on how media like newspapers, TV, and radio fit into your daily life. We are asking people we interview to give us extra cooperation and that they try hard to answer accurately so we can get complete and accurate information about this topic. You are one of the people we hope is willing to make the extra effort. Here is an agreement that explains what we are asking you to do. (HAND AGREEMENT) As you can see, it says, “I understand that the information from this interview must be very accurate in order to be useful. This means that I must do my best to give accurate and complete answers. I agree to do this.” We are asking people to sign an agreement and keep it for themselves so that we can be sure they understand what we are asking them to do. It is up to you to decide. If you are willing to agree to do this, we’d like you to sign your name here. (POINT OUT LINE) Down below there is a statement about confidentiality, and I will sign my name there. (POINT OUT LINE) (IF R HAS NOT ALREADY SIGNED) Are you willing to make the extra effort to continue the interview?

Priming

Priming is a concept in psychology to describe an automatic and unconscious process of exposure to one stimulus affecting responses to a later stimulus. Two studies have examined the use of a priming task to change respondents’ motivation to report honestly. Rasinski et al. (2004) gave undergraduate students a vocabulary task before asking them to complete a seemingly unrelated questionnaire on alcohol consumption. A random half of the respondents received neutral words; the other half received a combination of neutral words and words related to honesty. They found that respondents primed to be honest (through the vocabulary task with words related to honesty) reported more sensitive behaviors involving excessive alcohol consumption. Acquisti et al. (2012) primed a volunteer sample of *New York Times*

readers on privacy through a photo identification task. Participants were asked to identify either phishing emails or endangered fish before completing a Web survey. They found that the privacy priming through identification of phishing emails decreased respondents' propensity to admit socially undesirable behaviors (Study 2c). Question writers are encouraged to prime respondents' motivation to report truthfully.

Bogus Pipeline

The bogus pipeline technique aims to convince respondents that the researchers have a way of knowing if they lie. For example, Bauman and Dent (1982) warned respondents beforehand that their breath sample would be used to determine whether they had smoked before asking them about smoking. Tourangeau et al. (1997) told respondents that their inaccurate answers could be detected by a physiological recording device. Hanmer et al. (2014) told respondents through an introduction to the survey that their answers would be checked against public records kept by election officials. In all of these studies, the collection of additional information before survey administration was never used in reality to check against the truth, hence the name "bogus pipeline." All the studies reported that the technique effectively reduced social desirability bias in answers to sensitive questions.

Strategies on Writing Sensitive Questions

This section describes strategies on how to phrase sensitive questions, including manipulations on lead-ins or introductions, question wording, and response options.

Desensitizing the Question

Desensitizing a target-sensitive question by preceding it with an even more sensitive question has been found to elicit more disclosures of socially undesirable answers. For instance, Acquisti et al. (2012) empirically showed that significantly more respondents reported littering in a public space when the question was asked after sensitive questions on one's sexual life such as having an affair with a friend's spouse than when the question was preceded by less intrusive questions (e.g., failing to tip a waiter, failing to do chores). This is because a question becomes less sensitive and threatening when it is preceded by a more sensitive and threatening question. Although Bradburn et al. (2004) listed this approach as a potential strategy, they also acknowledged that this approach has limitations because it may lead to survey breakoffs. We suggest that question writers consider this approach with caution.

Choosing the Appropriate Time Frame

For socially undesirable behaviors, Bradburn et al. (2004) suggest starting with a question asking about "ever" before asking about "current" status. This is because questions about events that happened in the past are less salient and threatening than questions about current behaviors. The PATH Study adopts this approach by first asking if one ever smoked a cigarette, and for those answering yes, whether they now smoke every day, some days, or not at all (<https://www.icpsr.umich.edu/files/NAHDAP/pathstudy/36231-1001-Questionnaire-English.pdf>). In contrast, for socially desirable behaviors, Bradburn et al. (2004) suggest first asking about "current" behavior before asking about "ever."

Using Forgiving Introduction

Forgiving introductions, also called “face-saving,” or “permissive” introductions, are recommended when asking sensitive questions (Bradburn et al., 2004; Tourangeau & Yan, 2007). They are intended to reduce perceived sensitivity of the questions by normalizing or justifying the undesirable behavior or attitude and to eliminate any concerns about negative consequences of giving a truthful answer.

Question writers can utilize question order, time frame, forgiving introduction, question wording, response options, and indirect methods to encourage truthful response and to reduce misreporting to sensitive questions.

Forgiving introductions take several forms and can be summarized into four main groups. The first is the “everyone does it” approach. A humorous exemplification of this approach is shown by Barton (1958) (“As you know, many people have been killing their wives these days. Do you happen to have killed yours?”). Holtgraves et al. (1997) presented a survey question on vandalism with this type of forgiving introduction: “Almost everyone has probably committed vandalism at one time or another.” The second form of forgiving introduction uses an authority to justify the behavior: “Many doctors now believe that moderate drinking of liquor helps to reduce the likelihood of heart attacks or strokes” (Bradburn et al., 2004). The third type of forgiving introduction shows two or more sides of a story using the “some... others...” format. A typical example is provided by Peter and Valkenburg (2011): “Some people use erotic or pornographic materials often, while others do this rarely or never.” The fourth type of forgiving introduction provides a face-saving excuse for not doing something socially desirable or for doing something socially undesirable. Holtgraves et al. (1997) provided an example of such a face-saving introduction to an item about global warming: “You may not have had enough time to learn about the GATT treaty because of a heavy load of school work.” All these studies have found that providing forgiving instructions increases reporting of socially undesirable behaviors.

Question Wording Strategies

Several strategies have been used to improve the wording of sensitive questions. The first strategy **assumes the socially undesirable behavior and asks about it** (Bradburn et al., 2004; Tourangeau & Yan, 2007). For instance, to measure the number of cigarettes people smoke each day, we can ask respondents directly: “How many cigarettes do you smoke each day?” The second strategy employs **face-saving question wording**. One example is provided in Holtgraves and colleagues (1997): “Have you had the opportunity to learn the details of the Clinton Health Care Plan?” The third strategy **uses long questions with words familiar** to respondents, especially when asking socially undesirable behaviors (Bradburn et al., 2004). For instance, Blair and colleagues (1997) first asked respondents to provide their own word for intoxication and then used that word in the question below on frequency of intoxication:

Occasionally, people drink on an empty stomach or drink a little too much and become (RESPONDENT'S WORD FOR INTOXICATION). In the past year, how often did you become (RESPONDENT'S WORD FOR INTOXICATION) while drinking any kind of alcoholic beverage?

This question was found to outperform the standard questionnaire item (“In the past year, how often did you become intoxicated while drinking any kind of alcoholic beverage?”).

Response Option Strategies

As discussed in the earlier chapters, respondents use response options to make inferences on the norm of the behaviors. A high-frequency scale (0, 1–4, 5–9, 10–49, 50–99, 100 or more) conveys a higher norm than a low-frequency scale (0, 1, 2, 3, 4, 5 or more). As a result, respondents were found to report more sexual partners when given the high-frequency scale than the low-frequency scale (Tourangeau & Smith, 1996). Furthermore, respondents reported more sexual partners when no response list was provided than when a low-frequency list was used (Tourangeau & Smith, 1996). Consequently, we recommend that question writers **use the open-ended question format or a high-frequency scale** when asking frequency of socially undesirable behaviors (Bradburn et al., 2004; Tourangeau & Yan, 2007; Yan, 2021; Yan & Cantor, 2019).

Face-saving response options are used to lower the threat of truthfully disclosing socially undesirable behaviors. Zeglovitis and Kritzinger (2014) provided two additional face-saving response options to a question asking about voter turnout besides the yes/no dichotomous response options: “I thought about voting this time but didn’t” and “I usually vote but didn’t this time.” The addition of face-saving response options significantly increased the percentage of respondents reporting not voting in the last election.

The **unfolding brackets technique** is recommended for sensitive questions asking about income, assets, and other financial information as discussed in Chapter 5. Figure 5.2 provides a great example of this technique. Yan et al. (2010) showed empirically the positive impact of using unfolding brackets to reduce missing data to income questions.

Using Indirect Questions

Answers to direct questions reveal respondents’ involvement in socially (un)desirable behaviors. For instance, a “Yes” answer to a survey question asking about having had an abortion in the last 5 years clearly reveals that the respondent has had an abortion. By contrast, indirect methods aim to protect the anonymity of respondents’ answers by breaking the link between answers and involvement in the socially (un)desirable behaviors.

There are several types of indirect methods. The **randomized response technique (RRT)**, originally proposed by Warner (1965), relies on a randomizing mechanism (e.g., asking the respondent to flip a coin or throw a die) to determine that the respondent is supposed answer the sensitive question. There are different ways to do it. Greenberg et al. (1969) described two approaches. The first approach is to ask respondents to agree or disagree with either a sensitive statement (e.g., I have had an abortion) or the complement statement (e.g., I have not had an abortion). The second approach uses an unrelated and nonsensitive statement instead of the complement statement (e.g., “I was born in North Carolina”). A third approach is to instruct the respondent to answer the sensitive question truthfully when the randomization results in a certain outcome (e.g., when the coin flip is tails), but to give a predetermined answer (such as always responding “No”) otherwise (e.g., Boruch, 1971; Fox & Tracy, 1984). Holbrook and Krosnick (2010) provided an application of using RRT to measure voter turnout. The randomizing mechanism provides protection for the respondent’s true status with

respect to the sensitive question because there is a stochastic uncertainty as to which question was answered. A related variation – **the cross-wise model** – presents both the sensitive and the nonsensitive question to respondents and provides two response options for respondents to choose from. The two response options are “my answers are the same for both questions” and “my answers are different for both questions” (Jann et al., 2012).

The **item count technique (ICT)** is another type of indirect method in which respondents are randomly assigned to receive a long list of questions including the sensitive question of interest, or the same list without the sensitive question (Droitcour et al., 1991; Miller, 1984). Respondents are then asked to provide the number of items from the list they answered positively. A respondent’s true value on the sensitive question is protected because the answer is an overall count variable of all “Yes” responses. The prevalence of the sensitive behavior is estimated by taking the difference in the counts between the two lists. Variations of this method include using two pairs of lists (e.g., Biemer & Brown, 2005) and **item sum technique** designed for continuous variables (Trappmann et al., 2014).

Two major disadvantages are common to indirect methods. First, a larger sample size is needed because of the randomization. Second, estimates at the aggregate level can be calculated, but individual responses are not known, making it impossible to analyze the data at the individual level. Methods have been developed for multivariate regression analysis of data obtained from the ICT, enabling researchers to assess how the probability of answering Yes to the sensitive question varies as a function of respondents’ characteristics (Imai, 2011), but the application is not straightforward and may not be applicable to other types of indirect methods.

Strategies for Writing Questions for Older and Younger Respondents

Older and younger people are two sensitive populations. There is a growing literature demonstrating that older and younger respondents differ in cognitive functioning and are differentially affected by questionnaire features during the survey response process (Schwarz et al., 1999). As a result, question writers should take into consideration the cognitive abilities of their target population.

Asking Older Respondents

It is established that aging is related to a decline in basic functions, such as attention and working memory capacity (Salthouse & Babcock, 1991) and higher level functioning such as speech and language abilities, reasoning,

Question writers should be mindful about declining cognitive functioning among the aging population. Survey questions should be simple and short. And older adults should be given more time to answer survey questions.

and decision-making (Kutschar & Weichbold, 2019). The decline in cognitive functioning poses challenges at every stage of the survey response process, making it harder for older respondents to answer survey questions (see a review by Knäuper et al., 2016). For instance, older respondents are more likely to encounter comprehension issues with negatively worded items and complex content (Fox et al., 2007). Older respondents encounter greater difficulty remembering where or when they experienced an event or

learned a fact, partially because of decline in both long- and short-term memories and partially because they are likely to have more to retrieve (Craik, 1999). Older respondents have also been found to have difficulties with judgment and mapping – they are more likely to provide missing data or “Don’t know” answers (e.g., Kutschar & Weichbold, 2019), imprecise answers (Andrews & Herzog, 1986), inadequate answers (Holbrook et al., 2006; Olson et al., 2018), and answers that cannot be easily coded into categories (Olson et al., 2018). Finally, older respondents are found to have difficulty using numeric response options (Fox et al., 2007) and bipolar scales (Fox et al., 2007; Krestar et al., 2012).

On the positive side, declining working memory capacity makes previously used information less accessible to older respondents. Consequently, older respondents are less likely to be affected by question order effects (Knäuper et al., 2007), as discussed in Chapter 4. However, when answering questions on less salient and less relevant topics, they are more likely to be affected by the order of response options (Knäuper, 1999) and the number and range of response options (Knäuper et al., 2004).

Given the difficulties older respondents have answering survey questions, we suggest writing simpler and shorter questions and using fewer response options. We also suggest allowing older respondents to take time to answer questions; several studies have demonstrated that older adults who take longer to answer questions provide better quality data than those who answer faster (e.g., Yan et al., 2015).

Asking Younger Respondents

Children’s cognitive development is closely related to age. Theories differ on age cutoffs when describing children’s cognitive development stages. We adopt the cutoffs used by de Leeuw et al. (2004). Children under the age of 7 have not developed the cognitive functioning required to answer survey questions. As a result, we do not recommend surveying this age group directly; instead, we suggest proxy reporting by parents and guardians, combined with special data collection techniques aimed for children, such as playing assessment, drawings, story completion and puzzle tasks, and observational studies.

Children between the ages of 7 and 12 are better with logical and systematic thinking but are still growing in terms of cognitive processing and memory capacity. At that age, respondents have been shown to have difficulties with vague words (de Leeuw et al., 2004), negatively phrased questions (Borgers et al., 2000), indirect questions (Borgers et al., 2000; Scott, 1997), ambiguous questions (Borgers & Hox, 2001; Scott, 1997), and questions with more response options (Borgers & Hox, 2001). Interestingly, children with 4–6 years of education had less missing data to sensitive questions but more missing data to questions that were coded as not sensitive by researchers (Borgers & Hox, 2001). Children in this age group have a short attention span, may satisfice when they find the survey uninteresting or too long, or provide socially desirable or acquiescing answers because they can be easily influenced (Borgers et al., 2000; de Leeuw et al., 2004). They do not have difficulty navigating grid questions on the web (Maitland et al., 2016) and produce data of sufficient reliability and validity for questions using emojis to illustrate response

Question writers should be mindful about the cognitive development of children and write questions appropriate for their age.

options (e.g., [Massey, 2022](#)). Thus, if surveying children between 7 and 12 years of age, we suggest using simple language and offering two to three response options. If response scales with more than two or three options are to be used, consider using emojis instead of verbally labeling the long response scales.

Children aged 13–17 have well-developed cognitive functioning, including formal thinking, negations, and logic, and are considered able to answer survey questions on their own ([Borgers et al., 2000](#); [de Leeuw et al., 2004](#)). Studies show that youth of this age group provide equally reliable answers to questions about tobacco and health as adults ([Tourangeau et al., 2019](#)). In addition, they have been able to navigate web instruments and provide insights on their interactions and experiences with the web survey and adults ([Yang et al., 2021](#)). However, the setting of interviews is important for youth of this age group. Studies have found that youth reported more risk behaviors when interviewed at schools than at home ([Brener et al., 2006](#); [Fendrich & Johnson, 2001](#)).

Youth of this age group provide more reliable answers to demographic questions, questions with yes/no response options, and questions using simple words ([Tourangeau et al., 2020b](#)). We suggest writing questions that are short, clear, and use simple words, and using shorter response options. To assist youth, we suggest providing a brief and clear introduction using child-friendly language and a bullet point list of simple instructions on what is expected from them ([O'Reilly et al., 2013](#)). We also suggest keeping data collection private if interviews are done at the home of youth.

Summary

Survey questions are sensitive because they are personal or intrusive, there is a disclosure risk, and they invoke a social norm. In many cases, the perceived sensitivity of a survey item depends on whether the respondent has experienced the behavior of interest. We reiterate below strategies question writers can use to alleviate item nonresponse and to reduce socially desirable responding.

Strategies pertaining to survey design protocol include the following:

- 1 Using a self-administered mode
- 2 Using a diary if appropriate
- 3 Collecting data in private and ensuring anonymity

Strategies on how to prepare respondents for the task of answering sensitive questions include the following:

- 1 Providing instructions, asking for commitment, asking for an honesty pledge
- 2 Priming
- 3 Bogus pipeline technique

Strategies on writing sensitive questions include the following:

- 1 Embedding the question
- 2 Choosing an appropriate time frame
- 3 Using a forgiving introduction

- 4 Implementing wording strategies such as using questions presupposing behavior, using long questions with familiar words, and using face-saving wording
- 5 Implementing response option strategies such as open-ended format or a high-frequency list for questions asking about frequency or numbers, using unfolding brackets for questions on income and assets, and providing face-saving options

For respondents aged 65 and above and respondents under the age of 18, we suggest using short questions and fewer response options, and allowing more time to answer.

Exercises

- 1 How does social desirability affect survey responses?
- 2 Describe two ways you can reduce the effects of social desirability when asking respondents about drug use.
- 3 How would you write survey questions asking older respondents and teenagers about their drug use?